NucNetPy

Release 1.0.0

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NucNetPy is a pure-Python nuclear reaction-network package. It reads JINA/libnucnet XML data, evaluates reaction rates, builds and integrates the abundance equations with stiff solvers, solves nuclear statistical and quasi-statistical equilibrium, and provides the analysis quantities used in the NucNet Tools workflows.

It is a clean reimplementation: it does not call or wrap the original C/C++ code, and NumPy and SciPy are its only mandatory dependencies.

Chapter 1

Installation

1.1 Requirements

Python 3.10 or later, with NumPy 1.22+ and SciPy 1.9+. Nothing else is required to use the package.

1.2 From a checkout

```
git clone https://github.com/mtalafha90/NucNetPy.git
cd NucNetPy
python -m pip install -e .
pytest -q
```

The test suite runs in about a second and should report all tests passing before you rely on any result.

1.3 Optional extras

Extra	Adds	For
`dev`	`pytest`	running the test suite
`hdf5`	`h5py`	HDF5 persistence
`plot`	`matplotlib`, `networkx`	plotting and graph export
`notebook`	`jupyter`, `pandas`, `matplotlib`	the tutorial notebooks
`docs`	`sphinx`, `furo`, `myst-parser`	building this manual

```
python -m pip install -e ".[dev,hdf5,plot,notebook]"
```

`pandas` is used only by the notebooks to display tables; the package itself never imports it.

1.4 Verifying the installation

```
import nucnetpy as nn
print(nn.__version__)
```

The command-line interface is installed alongside:

```
nucnetpy --help
```

1.5 Building this manual

```
python -m pip install -e ".[docs]"
sphinx-build -b html docs/manual docs/manual/_build/html
```

For a single self-contained file, suitable for a submission archive:

```
sphinx-build -b singlehtml docs/manual docs/manual/_build/singlehtml
```

The PDF is written by rinohtype, without LaTeX:

```
make -C docs/manual pdf          # docs/manual/_build/pdf/nucnetpy.pdf
```

rinohtype typesets equations with the `ziamath` and `cairosvg` packages, which the `docs` extra installs. `cairosvg` also needs the Cairo library (`libcairo2` on Debian and Ubuntu, `brew install cairo` on macOS). Without them the PDF is still written, but every equation is left out and the build prints only a warning.

Chapter 2

Concepts

Five objects carry everything.

2.1 Species

A nuclide: name, charge `z`, mass number `a`, mass excess, ground-state spin, and an optional partition-function table.

```
import nucnetpy as nn

ni56 = nn.Species("ni56", z=28, a=56, mass_excess=-53.907, spin=0.0)
```

Mass excess and spin are not decoration. Equilibrium, detailed balance and energy release are all computed from mass excesses, and the statistical weight in an equilibrium solve is `(2J+1) * G(T)`, so a placeholder value produces a placeholder answer. A species whose spin is unknown is given a weight of one rather than a guess.

`Species.parse` builds one from a name alone, normalising aliases (`p` → `h1`, `alpha` → `he4`), but it cannot invent nuclear data.

2.1.1 Participants that are not nuclides

`gamma`, `electron`, `positron` and the neutrinos appear in reaction records so that charge and lepton number balance, but they carry no baryon number and are never part of the abundance vector. `nucnetpy.species.is_massless` identifies them. See Pitfalls.

2.2 Reaction

Reactants, products, a rate, a `Q`-value, and metadata. The rate may be a ReacLib seven-coefficient fit, a tabulated rate, a constant, or a callable; a reaction may carry more than one, and they add.

```
r = nn.Reaction.from_names(["c12", "he4"], ["o16", "gamma"],
                           rate_fits=[nn.RateFit([10, 0, 0, 0, 0, 0, 0])],
                           q_value=7.162)
```

The flow of a reaction is

$$F_r = \frac{\lambda_r \, \rho^{\max(n_r - 1, 0)}}{\prod_i m_{ir}!} \prod_i Y_i^{m_{ir}}, \tag{2.1}$$

where the products run over the reactants that carry baryon number, and n_r is that reactant order. `Reaction.nuclear_reactants` is the list actually used.

2.3 ReactionNetwork

A collection of reactions, with duplicate removal, filtering, and the construction of rates, flows, and dY_i/dt.

2.4 Zone

One composition together with its thermodynamic properties. Abundances are molar: `Y` is moles per gram, and the mass fraction is `X = A*Y`. A normalised composition has `sum(X) = 1`.

```
zone = nn.Zone(abundances={"si28": 1.0 / 28.0},
               properties={"t9": "5.0", "rho": "1e8"})
```

2.5 Network

Species, reactions and zones in one container, plus `validate()`, which reports species named by a reaction but absent from the species map, reactions that do not balance, and species carrying no nuclear data.

2.6 Units

Quantity	Unit
Temperature `t9`	10^9 K
Density `rho`	g cm^{-3}
Abundance `Y`	mol g^{-1}
Mass excess	MeV
`Q`-value	MeV
Rate	s^{-1} scaled by the density powers of the flow
Energy generation	erg g^{-1} s^{-1}
Time	s

Chapter 3

Common workflows

3.1 Reading a JINA/libnucnet database

```
import nucnetpy as nn

net = nn.read_jina_xml("nuclides.xml", "reaction_data.xml", zones_xml="zone.xml")
report = net.validate()
print(len(net.species), "species in memory,",
      len(net.reactions.reactions), "reactions")
print("without nuclear data:", len(report["species_without_nuclear_data"]))
```

Count carefully here. `len(net.species)` is the size of the in-memory network after parsing, not the number of entries in the nuclide file. Anything named by a reaction record with no entry in that file gets a synthesised placeholder, so it is present in memory whether or not the file described it.

On the JINA database used throughout this manual the nuclide file holds 7852 nuclides, while `len(net.species)` is 8131: those 7852 plus 279 placeholders. `validate()` reports no missing species precisely because the placeholders already exist, so `species_without_nuclear_data` is the field to read, not `missing_species`.

The two containers also differ. `net.species` holds nuclides only, while `net.species_names()` returns 8136 — the same 8131 plus the massless reaction participants `gamma`, `electron`, `positron` and the two neutrinos, which balance a reaction record but never enter the abundance vector.

Reading the database takes 20 to 30 s and holds it in about 0.9 GB, for a process peak near 1 GB. That is the practical ceiling for a pure-Python implementation, which is why the next step matters.

3.2 Cutting a network down to size

A one-zone calculation rarely needs the whole chart of nuclides.

```
from nucnetpy.network_limiter import select_species, limit_network

limit_network(net, select_species(net, zmax=20, amax=44))
```

`limit_network` keeps every reaction whose nuclides all survive the cut. Photons and leptons never decide the outcome, so the photodisintegrations stay; listing `"gamma"` is harmless. See Pitfalls for why this matters.

3.3 Evolving one zone

```
result = nn.evolve_zone(
    net, net.zone(0), nn.time_grid(0.0, 10.0, 120),
    thermo=nn.constant_thermo(t9=5.0, rho=1.0e8),
    method="bdf", rtol=1e-8, atol=1e-14,
)
assert result.success, result.message
```

`thermo` is any callable `(t, abundances) -> (T9, rho)`, so a prescribed temperature and density history drops straight in; `nucnetpy.hydro.Trajectory` wraps a tabulated one with interpolation.

Supported methods are `bdf`, `radau`, `lsoda`, `rk45` and `dop853`, which call SciPy; `adaptive_implicit_euler`; and the fixed-step `rk4`, `euler` and `implicit_euler` fallbacks. Reaction networks are stiff, so `bdf`, `radau`, `lsoda` and `adaptive_implicit_euler` are the practical choices.

`adaptive_implicit_euler` is the scheme of NucNet Tools: implicit Euler steps solved by Newton–Raphson iterations, each sized so that no abundance whose mass fraction exceeds `min_mass_fraction` (10^{-15}) changes by more than `max_change` (0.5 per cent). It is first-order accurate, so it takes more steps than BDF for the same accuracy. But it cannot turn a decaying abundance negative, so it keeps working where BDF fails as a fuel runs out in a large network (see Pitfalls):

```
result = nn.evolve_zone(net, zone, times, thermo=thermo,
                        method="adaptive_implicit_euler", max_change=0.005)
print(result.steps, result.rejected_steps)
```

3.3.1 Choosing the Jacobian

The flows are monomials in the abundances, so the Jacobian is available in closed form and costs one sweep over the reaction list, against `N+1` right-hand-side evaluations for finite differences.

```
nn.evolve_zone(..., jac_mode="analytic")    # default; >100x faster at 100 species
nn.evolve_zone(..., jac_mode="numerical")   # finite differences
nn.evolve_zone(..., jac_mode="sparsity")    # SciPy estimates it from the pattern
```

SciPy applies a sparsity pattern only when no Jacobian callable is supplied, so the pattern and an explicit Jacobian are alternatives rather than complements.

3.4 Equilibrium

```
nse = nn.solve_nse(net, t9=5.0, rho=1.0e8, ye=0.5)
print(nse.success, nse.xsum, nse.computed_ye)
```

The solve shares no machinery with the time integration, which is what makes it a genuine check on a network calculation — provided the network is running on the rate library's own reverse rates. See the caveat under *Thermodynamic consistency* below. Species carrying no nuclear data are excluded automatically; pass `require_nuclear_data=False` to override that, and read Pitfalls before doing so.

Constrained cluster equilibria (the libnuceq workflow) use `solve_qse` with `QSECluster` constraints. Plasma corrections are available through `nucnetpy.coulomb.nse_correction`.

3.5 Thermodynamic consistency

A network relaxes to the equilibrium of the rates it is given. ReacLib derives each reverse rate from its forward rate with the ground-state spins only, and the XML readers restore the partition functions it leaves out (`nn.apply_reverse_partition_functions`); without them the JINA alpha chain at $T_9 = 5$ settles a median 4.5 per cent away from NSE, with them 0.34 per cent. What remains comes from the library's Q values and fits.

```
consistent = nn.consistent_reverse_network(net)
```

This pairs each reaction with its inverse, keeps the exothermic member (judged from the masses), and regenerates the other from detailed balance. On the JINA alpha chain it brings the integrated composition and a separately solved NSE to 8 parts in 10^8, once the equilibrium is solved over the same species set the network can populate.

Read that second figure carefully. The regenerated reverse rates come from the same equilibrium prefactor `solve_nse` uses, so the two calculations now share their nuclear data: an error in the masses, the `(2J+1)` factor or the partition-function convention would move both together and cancel. That agreement measures how consistently the integrator, the stoichiometry and the equilibrium solver treat one shared formulation, not whether that formulation is right.

It is an option rather than a default, because it replaces measured reverse rates with values derived from mass differences. Pass `tabulate=True` if the network must remain writable to XML, at some cost in accuracy.

3.6 Screening

```
from nucnetpy import SkyNetScreening

nn.evolve_zone(..., screening=SkyNetScreening(net.species))
```

Each nuclear charge is assigned a Coulomb chemical potential `mu(Z)/kT`, and a reaction is enhanced by the difference between the separated reactants and the fused compound charge. Being a chemical potential, it applies consistently to forward and reverse rates, so screening does not undo detailed balance. A pairwise Salpeter factor is also available through `reaction_screening_factor`.

The electrons screen too, by an amount that depends on how degenerate they are, and `SkyNetScreening` works it out from the electron gas (`nucnetpy.electrons`):

- for non-degenerate electrons, as in hydrogen and helium burning in ordinary stars, it recovers the Salpeter weak-screening limit;
- for degenerate electrons, as in a white dwarf, they no longer screen;
- in hot, thin matter, positron pairs add to the screening.

The electron state is recomputed only when the temperature, density or `Ye` changes, which costs about a millisecond. To fix the electrons' share yourself, pass a number as `pair_term`: `pair_term=0.0` keeps the ions only, and `pair_term=Ye` treats the electrons as non-degenerate.

3.7 Entropy

```
s = nn.entropy_per_nucleon(zone.abundances, net, t9=5.0, rho=1.0e8)
s.ions, s.electrons, s.photons, s.coulomb, s.total
```

The entropy per nucleon, in units of Boltzmann's constant, comes in parts:

- **ions** — the Sackur-Tetrode entropy of each species, with its real mass, ground-state spin and partition function (the last adds the entropy of the excited states);
- **electrons and positrons** — from the electron gas, at any degeneracy;
- **photons** — black-body radiation;
- **Coulomb** — the plasma correction, off unless `coulomb=True`.

To go the other way, `nn.density_for_entropy(s, abundances, net, t9)` and `nn.t9_for_entropy(s, abundances, net, rho)` find the density or temperature that gives entropy `s` at fixed composition. The entropy changes steadily with both, so the answer is unique.

`nn.constant_entropy_thermo(s, density, net)` turns this into a `thermo` callable: you give the density as a function of time, and the temperature is set at each step so that the current composition has entropy `s`. This is the usual way to follow matter as it expands and cools. The heat from the reactions is not fed back, so use it once the burning has slowed, as in a freeze-out. Burning that reduces the number of particles (helium to carbon, say) makes the temperature rise at fixed entropy.

3.8 Energy release

```
from nucnetpy import nuclear_energy_generation_rate, nuclear_energy_release
```

The rate follows from the change in total mass excess, `eps = -N_A * C * sum_i (dY_i/dt) * dM_i`, where `C = 1.602176634e-6 erg/MeV` converts the mass excesses from MeV. It does not use the reaction `Q`-values stored in the rate library, so it is internally consistent with the adopted mass-excess table, the same one `solve_nse` uses. It may differ from library `Q`-values when the two data sources use different nuclear masses. Neutrino losses are not subtracted.

3.9 Analysis

`nucnetpy.analysis` carries the diagnostics of the NucNet Tools workflows: largest mass fractions, element abundances, abundance moments, per-species timescales `Y/|dY/dt|`, per-reaction contributions to `dYe/dt`, entropy generation, `s`-process neutron exposure, integrated reaction currents, and separation energies.

3.10 Command line

Every operation above has a command-line equivalent; see Command line.

Chapter 4

Pitfalls

Everything in this chapter produces a plausible-looking answer rather than an error message. Each item was found by running the package against a production database rather than a test network, and each is now covered by a regression test — but the underlying trap belongs to the problem, not to this implementation, so it is worth knowing whichever code you use.

4.1 Dropping the photon when you cut a network

A code that keeps a reaction only if *all* its participants survive a cut, and cuts by a list of nuclides, discards every photodisintegration, because the photon is not a nuclide. Nothing warns you. The network simply cannot reach equilibrium, and a silicon-burning calculation will not start at all, because it begins with $^{28}\mathrm{Si}(\gamma,\ \alpha)^{24}\mathrm{Mg}$.

`limit_network` avoids this: only nuclides decide whether a reaction survives, so photons and leptons are kept automatically and listing `"gamma"` is harmless. Development versions of NucNetPy before the 1.0.0 release fell into the trap too, and with `include_linked=True` they also treated the photon as a link between otherwise unrelated reactions:

```
limit_network(net, select_species(net, zmax=20))    # keeps
the photodisintegrations
```

4.2 Using library reverse rates without their partition functions

A network relaxes to the equilibrium implied by the **ratios of the rates it is given**, not to the equilibrium of the nuclear data. ReacLib derives each reverse rate from its forward rate by detailed balance with the ground-state spins only, and expects the user to multiply it by the ratio of partition functions, $\prod G_{\mathrm{products}} / \prod G_{\mathrm{reactants}}$ of the reverse reaction.

Left out, that factor is not small. On the alpha chain at $T_9 = 5$ the reverse fits then differ from detailed balance by a median factor of 1.10 and up to 1.72 (for $^{56}\mathrm{Ni}(\gamma,\ \alpha)$), and the network relaxes to a composition a median 4.5 per cent and up to 48 per cent away from NSE. Versions of NucNetPy before the fix did exactly that; the comparison with `solve_nse` is what exposed it.

The XML readers now recognise the derived reverse rates from their coefficients, because libnucnet files do not usually carry ReacLib's `v` flag, and apply the ratio: the alpha chain then matches NSE to a median 0.34 per cent. For a network built in memory, call `apply_reverse_partition_functions(net)` yourself.

The coefficients show which two rates form a pair, but not which of the two was derived. The reader takes ReacLib's `v` flag from the fit labels when they carry it. Otherwise it takes the photodisintegration of a capture/photodisintegration pair, and the endothermic member of any other pair. Choosing the endothermic member everywhere, as NucNetPy 1.0.0 did, put the factor on the wrong rate in 1,862 of the 32,676 pairs of the ReacLib snapshot of 29 March 2022. Those are captures into nuclei

beyond the drip lines, which are endothermic, yet ReacLib still fits the capture and derives the photodisintegration. Eighteen two-body pairs in that snapshot can be told apart only by the flag. Write the labels with the flag (for example `ls09rv`) if you build a libnucnet file from ReacLib yourself.

What is left comes from the library's Q values and the accuracy of its fits. Use `consistent_reverse_network` if you need the network and the equilibrium solver to agree exactly, and understand that you are then trusting the nuclide file's masses over the library's Q values.

4.3 Electron captures without the electron density

ReacLib fits its three electron captures (on ^{3}He and ^{7}Be, and the pep reaction p + p + e^{-}) per unit electron density, ρY_e. Treated like a photodisintegration, whose rate already contains the photon density, they run too slowly by that factor: by about 3 in the core of a massive star on the main sequence, and by about 100 in the Sun. NucNetPy multiplies the ReacLib fits of a reaction by ρY_e once for each electron among its reactants. `Reaction.rate` and `ReactionNetwork.rates` need `ye` for that. `Reaction.flux`, `ReactionNetwork.flows`, `ReactionNetwork.ydot` and the network's right-hand side work it out from the abundances. NucNetPy 1.0.0 left the factor out.

4.4 Running out of fuel with BDF

As a fuel runs out, the higher-order BDF formula can predict a slightly negative abundance for it. The right-hand side treats a negative abundance as zero, which switches off every reaction that consumes the fuel, and SciPy's step size then collapses. On an 891-nuclide network from hydrogen to nickel this stopped hydrogen burning soon after 1.8 of its 6.7 Myr. Neither removing the clipping nor making the Jacobian agree with it cured the failure: each let the integration go further and then fail in a later burning stage.

`method="adaptive_implicit_euler"` avoids the problem. It takes implicit Euler steps, which cannot turn a decaying abundance negative, as NucNet Tools does, and sizes each step so that no abundance changes by more than `max_change` (0.5 per cent by default). It is first-order accurate: halving `max_change` halves the error. Use it when a large network runs a fuel down.

4.5 Placeholder nuclear data in an equilibrium solve

A production reaction file names nuclides that its nuclide file does not describe — 279 of them in the database used for the manuscript. NucNetPy synthesises placeholders so the reactions remain usable, with a mass excess of zero.

A mass excess of zero is not a neutral default: it makes an unbound nuclide look as tightly bound as the most stable one. Left in an NSE solve, ^{5}Li reached a mass fraction of 0.28 and displaced the iron peak. They are therefore recorded in `network.validate()["species_without_nuclear_data"]` and excluded from equilibrium solves by default.

4.6 Trusting `success` without checking it

```
result = nn.evolve_zone(...)
assert result.success, result.message
```

`success` is false when the solver failed, and also when positivity projection had to invent enough baryon number to break conservation by more than the requested tolerance.

That second case is worth understanding. The right-hand side clips its input to non-negative values, so once a component goes negative the derivative stops depending on it and its Jacobian column

vanishes; an implicit solver is then free to carry that component far negative while still passing its own convergence test. Clipping it back to zero at the end converts a large negative number into invented mass. In one long integration this returned $\sum_i X_i = 8.56$ with $X(^4\mathrm{He}) = 7.18$, while SciPy reported that it had reached the end of the interval successfully.

Checking conservation costs one inner product per output step. Do it.

4.7 Forgetting the spin degeneracy

libnucnet partition-function tables are *normalised*: they tend to one as $T \to 0$ for every nuclide, high-spin ones included. The statistical weight is $(2J + 1)\, G(T)$, and the degeneracy has to be supplied separately. Omitting it makes every nuclide with $J \neq 0$ too rare by that factor — up to 90 per cent for ^{50}V.

NucNetPy applies it automatically when a species carries a `spin`. If your species were built with `Species.parse`, they do not.

4.8 Tolerances

The default `atol` is an absolute floor on abundances. Set it far below the smallest abundance you care about, but not so far that the solver chases numerical noise in species that are effectively absent — that is a common cause of a stiff solve failing to start at all. Confirm the answer has stopped moving:

```
for rtol in [1e-4, 1e-6, 1e-8, 1e-10]:
    r = nn.evolve_zone(..., rtol=rtol, atol=rtol * 1e-8)
```

4.9 Interpolating a detailed-balance reverse rate

The reverse rate carries $\exp(-Q/kT)$, which is far too steep to interpolate. Tabulating it on 200 logarithmically spaced points costs three orders of magnitude of accuracy against evaluating it exactly. `consistent_reverse_network` attaches a callable by default for this reason; `tabulate=True` is for when the network must be serialisable.

Chapter 5

Verifying a calculation

Two different questions are worth keeping apart.

Verification asks whether the code does what it was designed to do. *Validation* asks whether that design describes nature, and needs a reference outside the code.

5.1 What the package checks for you

The test suite covers XML round-tripping as a fixed point, ReacLib rates on a T_9 grid, screening factors, weak-rate interpolation, right-hand-side values and their baryon residual, full trajectories, the analytic Jacobian against finite differences, the treatment of photons and leptons, solver dispatch, the spin degeneracy, exclusion of species without nuclear data, and detection of a positivity projection that manufactures mass.

```
pytest -q
```

5.2 What you should check on your own calculation

Conservation. For a reaction set that conserves baryon number, the instantaneous residual

$$\delta_A(t) = \sum_i A_i \frac{dY_i}{dt} \tag{5.1}$$

should stay at round-off, and $\sum_i A_i Y_i$ should stay at its initial value. Track them along the whole trajectory, not only at the start.

These are two different quantities and are worth keeping apart when you quote them. δ_A is a **rate**, in s^{-1}: how fast the right-hand side fails to conserve baryon number at one instant. The departure of $\sum_i A_i Y_i$ from unity is a **dimensionless** normalisation error of the composition itself. A third quantity — the baryon number a positivity projection had to manufacture, reported when a solve fails — is a dimensionless fraction of the initial amount, different again. On a well-behaved problem all three sit near round-off, which makes them easy to conflate.

Solver agreement. Repeat the calculation with `bdf`, `radau` and `lsoda`. They should agree far more closely than any physical effect you care about.

Tolerance convergence. Tighten `rtol` and confirm the endpoint stops moving.

Equilibrium. Where the physics allows it, run the calculation to its stationary state and compare with `solve_nse`. This is the strongest check available, but what it tests depends on where the reverse rates came from, and the two cases are worth keeping apart.

With the library's own reverse fits the two sides share no numerical input, so a disagreement is informative. It exposed reverse rates evaluated without their partition functions, a defect now fixed (see Pitfalls on reverse rates); what remains, about 0.3 per cent on the alpha chain at $T_9 = 5$, measures

the library's Q values and fits against the nuclide file.

With reverse rates rebuilt by `consistent_reverse_network` the two sides are no longer independent: the reverse rates are derived from the same equilibrium prefactor `solve_nse` uses, so the masses, the `(2J+1)` factor and the partition-function convention are common to both and cancel from the comparison. That agreement is a strong end-to-end test of the integrator, the stoichiometry and the equilibrium solver's root finding, and says nothing about whether the shared formulation is right. For that you need a reference outside this package.

5.3 Golden files and what they do not prove

`tests/golden/` holds frozen outputs, and `nucnetpy.validation.regression_summary` compares against them. These detect accidental changes in behaviour. They are snapshots of this implementation, so they are not independent evidence that an answer is right, and they must not be described as agreement with the original C++ code.

`validation/generate_golden.py` writes the reference records in the format the suite consumes, so outputs from a NucNet Tools build can be substituted for the self-generated ones without changing any test code.

5.4 Reproducing the published results

```
python validation/demonstration.py \
    --nuclides nuclides.xml --reactions reaction_data.xml --zone zone.xml \
    --outdir results/

python validation/benchmark_performance.py \
    --nuclides nuclides.xml --reactions reaction_data.xml \
    --json results/benchmark.json --latex results/table_benchmark.tex
```

Both record the interpreter, NumPy, SciPy and platform versions alongside their output.

Chapter 6

Python API

Everything below is exported at the top level of `nucnetpy` unless noted. Pages are generated from the docstrings in the source.

6.1 Core objects

Nuclide/species utilities.

This module provides the most frequently needed pieces of libnucnet species handling in pure Python: name parsing, Z/A/N bookkeeping, mass fractions, charge fractions, and common element symbols.

nucnetpy.species.**MASSLESS_SPECIES = frozenset({'anti-neutrino_e', 'antineutrinoe', 'electron', 'gamma', 'neutrino_e', 'neutrinoe', 'positron'})**

Reaction participants that carry no baryon number. Photons and leptons appear explicitly in JINA/libnucnet reaction records so that charge and lepton number can be checked, but they are not part of the abundance vector: the photon bath and the electron sea are properties of the thermodynamic state, and their densities are already folded into the tabulated rate. They must therefore be excluded from the abundance product, the reactant order, and the identical-particle statistical factor of a reaction flow.

nucnetpy.species.**is_massless** (*name*)

Return True for photons and leptons, which carry no baryon number.

Parameters **name** (*str*)

Return type bool

class nucnetpy.species.**Species** (*name*, *z*, *a*, *mass_excess=0.0*, *spin=None*, *source=None*, *partition=<factory>*)

Bases: object

A nuclear species.

Parameters are intentionally similar to libnucnet's Nuc objects. *mass_excess* is in MeV when available. *partition* may store temperature dependent partition-function data.

Parameters

- **name** (*str*)
- **z** (*int*)
- **a** (*int*)
- **mass_excess** (*float*)
- **spin** (*float | None*)
- **source** (*str | None*)
- **partition** (*Dict[float, float]*)

name: str

z: int

a: int

mass_excess: float = 0.0

spin: float | None = None

source: str | None = None

partition: Dict[float, float]

property n: int

property element: str

property latex: str

classmethod parse (*value*, ***kwargs*)

Parameters **value** (*str*)

Return type *Species*

nucnetpy.species.**normalize_species_name** (*value*)

Parameters **value** (*str*)

Return type str

nucnetpy.species.**species_from_za** (*z*, *a*, ***kwargs*)

Parameters
- **z** (*int*)
- **a** (*int*)

Return type *Species*

nucnetpy.species.**mass_fraction** (*y*, *a*)

Parameters
- **y** (*float*)
- **a** (*int*)

Return type float

nucnetpy.species.**abundance_from_mass_fraction** (*x*, *a*)

Parameters
- **x** (*float*)
- **a** (*int*)

Return type float

Core containers: Network, Zone, and multi-zone collections.

class nucnetpy.core.**Zone** (*label: 'Tuple[str, str, str]'=('0', '0', '0'), abundances: 'Dict[str, float]'=<factory>, properties: 'Dict[str, str]'=<factory>, optional_properties: 'Dict[str, str]'=<factory>*)

Bases: object

Parameters
- **label** (*Tuple[str, str, str]*)
- **abundances** (*Dict[str, float]*)

- **properties** (*Dict[str, str]*)
- **optional_properties** (*Dict[str, str]*)

label: **Tuple**[**str**, **str**, **str**] = ('0', '0', '0')

abundances: **Dict**[**str**, **float**]

properties: **Dict**[**str**, **str**]

optional_properties: **Dict**[**str**, **str**]

set_abundance (*species*, *y*)

Parameters
- **species** (*str*)
- **y** (*float*)

Return type None

get_abundance (*species*, *default=0.0*)

Parameters
- **species** (*str*)
- **default** (*float*)

Return type float

mass_fractions (*species_map=None*)

Parameters **species_map** (*Mapping[str, Species] | None*)

Return type *Dict*[str, float]

ye (*species_map=None*)

Parameters **species_map** (*Mapping[str, Species] | None*)

Return type float

ysum ( )

Return type float

xsum (*species_map=None*)

Parameters **species_map** (*Mapping[str, Species] | None*)

Return type float

temperature9 (*default=1.0*)

Parameters **default** (*float*)

Return type float

density (*default=1.0*)

Parameters **default** (*float*)

Return type float

class nucnetpy.core.**Network** (*species: 'Dict[str, Species]'=<factory>, reactions: 'ReactionNetwork' = <factory>, zones: 'List[Zone]' = <factory>, metadata: 'Dict[str, str]'=<factory>*)

Bases: object

Parameters
- **species** (*Dict[str, Species]*)

- **reactions** (*ReactionNetwork*)
- **zones** (*List[Zone]*)
- **metadata** (*Dict[str, str]*)

species: Dict[str, Species]

reactions: ReactionNetwork

zones: List[Zone]

metadata: Dict[str, str]

add_species (*species*)

Parameters **species** (*Species*)

Return type None

ensure_species (*name*)

Parameters **name** (*str*)

Return type *Species*

add_zone (*zone*)

Parameters **zone** (*Zone*)

Return type None

species_names ( )

Return type *List*[str]

zone (*index=0*)

Parameters **index** (*int*)

Return type *Zone*

validate ( )

Return type *Dict*[str, object]

species_without_nuclear_data ( )

Return species that were synthesised to satisfy a reaction record.

These are referenced by the reaction file but absent from the nuclide file, so they carry a placeholder mass excess of zero rather than measured or evaluated data. They are safe for structural operations such as counting or plotting the network, but must not enter an equilibrium calculation, where a mass excess sets the binding energy.

Return type *List*[str]

abundance_matrix (*species=None*)

Parameters **species** (*Sequence[str] | None*)

Return type *Tuple*[*List*[str], *ndarray*]

Reaction and rate machinery for a pure-Python NucNet Tools replacement.

The implementation supports common ReacLib 7-coefficient rates, tabulated rates, manual constants, duplicate handling, conservation checks, and stoichiometric flux evaluation suitable for one-zone evolution and analysis commands.

class nucnetpy.reactions.**ReactionParticipant** (*species: 'str', count: 'int' = 1*)

Bases: object

Parameters
- **species** (*str*)
- **count** (*int*)

species: str

count: int = 1

class nucnetpy.reactions.**RateFit** (*coefficients, label='', chapter=None*)

Bases: object

A standard seven-parameter ReacLib rate fit.

lambda = exp(a0 + a1/T9 + a2/T9^(1/3) + a3*T9^(1/3) + a4*T9

- a5*T9^(5/3) + a6*ln(T9))

A reaction may carry several such fits, and `Reaction.bare_rate` sums them. The result is a generic rate coefficient, not N_A<sigma v>: the same form serves one-body decays and photo-disintegrations, two-body captures and three-body reactions, so its units depend on the number of nuclear reactants. The density factor in `Reaction.flux` converts it to an abundance flow.

Parameters
- **coefficients** (*Sequence[float]*)
- **label** (*str*)
- **chapter** (*int | None*)

coefficients: Sequence[float]

label: str = ''

chapter: int | None = None

rate (*t9*)

Parameters **t9** (*float*)

Return type float

class nucnetpy.reactions.**PartitionRatio** (*numerator=(), denominator=()*)

Bases: object

The factor `prod G(products) / prod G(reactants)` of a reverse rate.

ReacLib derives each reverse rate (flag `v`) from its forward rate by detailed balance with the ground-state spins only, and expects the user to multiply it by this ratio of normalised partition functions. Each entry is `(t9 points, G values, multiplicity)` for one nuclear species, taken from the nuclide file and interpolated exactly as the NSE solver interpolates it, so that a corrected reverse rate and `nucnetpy.solve_nse()` use the same partition functions. See `nucnetpy.apply_reverse_partition_functions()`.

Parameters
- **numerator** (*Tuple[Tuple[Tuple[float, ...], Tuple[float, ...], int], ...]*)
- **denominator** (*Tuple[Tuple[Tuple[float, ...], Tuple[float, ...], int], ...]*)

numerator: Tuple[Tuple[Tuple[float, ...], Tuple[float, ...], int], ...] = ()

denominator: Tuple[Tuple[Tuple[float, ...], Tuple[float, ...], int], ...] = ()

class nucnetpy.reactions.**TabularRate** (*t9: 'Sequence[float]', rates: 'Sequence[float]', log_interp: 'bool' = True*)

Bases: object

Parameters
- **t9** (*Sequence[float]*)
- **rates** (*Sequence[float]*)
- **log_interp** (*bool*)

t9: Sequence[float]

rates: Sequence[float]

log_interp: bool = True

rate (*t9*)

Parameters **t9** (*float*)

Return type float

class nucnetpy.reactions.**Reaction** (*reactants: 'List[ReactionParticipant]', products: 'List[ReactionParticipant]', rate_fits: 'List[RateFit]' = <factory>, tabular_rate: 'Optional[TabularRate]' = None, q_value: 'float' = 0.0, source: 'str' = '', label: 'str' = '', metadata: 'Dict[str, str]' = <factory>, constant_rate: 'Optional[float]' = None, rate_function: 'Optional[Callable[[float], float]]' = None, partition_ratio: 'Optional[PartitionRatio]' = None*)

Bases: object

Parameters
- **reactants** (*List[ReactionParticipant]*)
- **products** (*List[ReactionParticipant]*)
- **rate_fits** (*List[RateFit]*)
- **tabular_rate** (*TabularRate | None*)
- **q_value** (*float*)
- **source** (*str*)
- **label** (*str*)
- **metadata** (*Dict[str, str]*)
- **constant_rate** (*float | None*)
- **rate_function** (*Callable[[float], float] | None*)
- **partition_ratio** (*PartitionRatio | None*)

reactants: List[ReactionParticipant]

products: List[ReactionParticipant]

rate_fits: List[RateFit]

tabular_rate: TabularRate | None = None

q_value: float = 0.0

source: str = ''

label: str = ''

metadata: Dict[str, str]

constant_rate: float | None = None

rate_function: Callable[[float], float] | None = None
Optional `f(t9) -> rate` evaluated exactly at the requested temperature. It is used for rates that are computed rather than fitted or tabulated, such as the detailed-balance reverse rates built by `nucnetpy.consistent_reverse_network()`, whose steep exponential temperature dependence is poorly captured by interpolation. A callable cannot be written to XML, so a network containing one does not round-trip; tabulate the rate instead when persistence is required.

partition_ratio: PartitionRatio | None = None
Partition-function factor applied to the ReacLib fits of a library reverse rate, which the library fits with ground-state spins only. Set by `nucnetpy.apply_reverse_partition_functions()`, which the XML readers call; `None` for every other reaction.

classmethod from_names (*reactants*, *products*, ***kwargs*)

Parameters
- **reactants** (*Sequence[str]*)
- **products** (*Sequence[str]*)

Return type *Reaction*

property key: Tuple[Tuple[Tuple[str, int], ...], Tuple[Tuple[str, int], ...]]

property string: str

property latex: str

property nuclear_reactants: List[ReactionParticipant]
Reactants that carry baryon number.
Photons and leptons are dropped: a photodisintegration such as `si28 + gamma -> mg24 + he4` is a one-body process whose rate already contains the photon density, so including `gamma` in the abundance product would multiply every such flow by `Y(gamma) = 0` and would raise the density power by one.

property reactant_order: int

property electron_count: int
Number of electrons among the reactants; non-zero for an electron capture.

bare_rate (*t9*, *rho=1.0*, *ye=None*)
Return the rate coefficient before screening.
The ReacLib fits, the tabulated rate, the constant rate and the rate function are added together. Two factors apply to the ReacLib fits alone. The first is the partition-function ratio of a library reverse rate (`partition_ratio`). The second is the electron density of an electron capture: ReacLib fits its electron captures (on 3He and 7Be, and the pep reaction p + p + e-) per unit `rho * Ye` (Cyburt et al. 2010, ApJS 189, 240), so their fitted part is multiplied by `(rho * Ye) ** n` for `n` electron reactants. That factor needs `ye`; without it the fit is returned per unit electron density. The network's right-hand side and Jacobian always supply `Ye`.

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float | None*)

Return type float

rate (*t9*, *rho=1.0*, *ye=None*, *screening=None*)

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float | None*)
- **screening** (*Callable[[Reaction, float, float, float | None], float] | None*)

Return type float

statistical_factor ( )

Return type int

flux (*abundances*, *t9*, *rho=1.0*, *screening=None*, *ye=None*)

Return the flow of this reaction, in reactions per nucleon per second.
For an electron capture, `ye` defaults to the electron fraction of `abundances` (see `bare_rate()`).

Parameters
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)
- **screening** (*Callable[[Reaction, float, float, float | None], float] | None*)
- **ye** (*float | None*)

Return type float

stoichiometry ( )

Return type *Dict*[str, int]

conserves_a_z (*species_map*)

Parameters **species_map** (*Mapping[str, Species]*)

Return type *Tuple*[bool, int, int]

nucnetpy.reactions.**electron_fraction** (*abundances*)

Return `Ye = sum(Z * Y)` over the nuclides named in `abundances`.
Names that are not nuclides (photons, leptons) and negative abundances contribute nothing.

Parameters **abundances** (*Mapping[str, float]*)

Return type float

class nucnetpy.reactions.**ReactionNetwork** (*reactions: 'List[Reaction]' = <factory>*)

Bases: `object`

Parameters **reactions** (*List[Reaction]*)

reactions: **List**[**Reaction**]

add (*reaction*)

Parameters **reaction** (*Reaction*)

Return type None

extend (*reactions*)

Parameters **reactions** (*Iterable[Reaction]*)

Return type None

remove_duplicates ( )

Return type *ReactionNetwork*

invalid_reactions (*species_map*)

Parameters **species_map** (*Mapping[str, Species]*)

Return type *List*[*Reaction*]

filter_valid (*species_map*)

Parameters **species_map** (*Mapping[str, Species]*)

Return type *ReactionNetwork*

rates (*t9*, *rho=1.0*, *ye=None*)

Return the rate of every reaction.

Electron captures need `ye`; without it their rates are per unit electron density (see `Reaction.bare_rate()`).

Parameters

- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float | None*)

Return type *Dict*[str, float]

flows (*abundances*, *t9*, *rho=1.0*, *screening=None*, *ye=None*)

Return the flow of every reaction.

`ye` defaults to the electron fraction of `abundances`; electron captures need it (see `Reaction.bare_rate()`).

Parameters

- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)
- **screening** (*Callable[[Reaction, float, float, float | None], float] | None*)
- **ye** (*float | None*)

Return type *Dict*[str, float]

ydot (*abundances*, *t9*, *rho=1.0*, *screening=None*, *ye=None*)

Return `dY/dt` for every nuclide the reactions touch.

Photons and leptons are left out: they are not part of the abundance vector, so a `dY/dt` for them would be a meaningless number that downstream diagnostics would report as if it were a species. `ye` defaults to the electron fraction of `abundances`; electron captures need it (see `Reaction.bare_rate()`).

Parameters

- **abundances** (*Mapping[str, float]*)

- **t9** (*float*)
- **rho** (*float*)
- **screening** (*Callable[[Reaction, float, float, float | None], float] | None*)
- **ye** (*float | None*)

Return type *Dict*[str, float]

species_names ( )

Return type *List*[str]

class nucnetpy.reactions.**DetailedBalanceOptions** (*t9*, *rho*, *ye*, *equilibrium_ratio=1.0*)

Bases: object

Options for estimating reverse rates from forward rates.

Exact NucNet matching requires identical masses, partition functions, and Coulomb corrections. This helper provides a practical replacement hook.

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)
- **equilibrium_ratio** (*float*)

t9: **float**

rho: **float**

ye: **float**

equilibrium_ratio: **float** = **1.0**

nucnetpy.reactions.**reverse_rate_from_detailed_balance** (*forward_rate*, *equilibrium_ratio*)

Return reverse rate using `lambda_rev = lambda_fwd / K_eq`.

Parameters
- **forward_rate** (*float*)
- **equilibrium_ratio** (*float*)

Return type float

nucnetpy.reactions.**read_reaclib_text** (*text*)

Read a small ReacLib-like text block.

This parser accepts common two-line ReacLib records and a compact one-line form: `reactants -> products ; a0 ... a6`. It is intentionally permissive for migration from C++ examples and hand-written test networks.

Parameters **text** (*str*)

Return type *ReactionNetwork*

6.2 Evolution

One-zone and multi-zone evolution tools.

This layer implements the standard stoichiometric ODE interface and provides fixed-step, SciPy BDF

/Radau/LSODA, optional sparse Jacobians, positivity projection, screening hooks, and weak-rate source terms.

nucnetpy.solver.**METHODS = ('bdf', 'radau', 'lsoda', 'rk45', 'dop853', 'adaptive_implicit_euler', 'rk4', 'euler', 'implicit_euler', 'backward_euler')**

Every method name accepted by `evolve_zone()` (case-insensitive).

nucnetpy.solver.**constant_thermo** (*t9=1.0*, *rho=1.0*)

Parameters
- **t9** (*float*)
- **rho** (*float*)

Return type *Callable*[[float, *Mapping*[str, float]], *Tuple*[float, float]]

nucnetpy.solver.**zone_thermo** (*zone*)

Parameters **zone** (*Zone*)

Return type *Callable*[[float, *Mapping*[str, float]], *Tuple*[float, float]]

nucnetpy.solver.**time_grid** (*t0*, *t1*, *n*, *log=False*)

Parameters
- **t0** (*float*)
- **t1** (*float*)
- **n** (*int*)
- **log** (*bool*)

Return type *ndarray*

class nucnetpy.solver.**EvolutionResult** (*time: 'np.ndarray'*, *species: 'List[str]'*, *y: 'np.ndarray'*, *success: 'bool' = True*, *message: 'str' = ''*, *nfev: 'int' = 0*, *njev: 'int' = 0*, *steps: 'int' = 0*, *rejected_steps: 'int' = 0*)

Bases: `object`

Parameters
- **time** (*ndarray*)
- **species** (*List[str]*)
- **y** (*ndarray*)
- **success** (*bool*)
- **message** (*str*)
- **nfev** (*int*)
- **njev** (*int*)
- **steps** (*int*)
- **rejected_steps** (*int*)

time: ndarray

species: List[str]

y: ndarray

success: bool = True

message: str = ''

nfev: int = 0

njev: int = 0

steps: int = 0
Accepted and rejected steps; counted by `adaptive_implicit_euler` only.

rejected_steps: int = 0

property final_abundances: Dict[str, float]

zone ( )

Return type *Zone*

mass_fraction_history (*species_map=None*)

Return type *ndarray*

nucnetpy.solver.**rhs** (*network*, *species*, *thermo*, *screening=None*, *weak_rates=None*)

Parameters
- **network** (*Network*)
- **species** (*Sequence[str]*)
- **thermo** (*Callable[[float, Mapping[str, float]], Tuple[float, float]]*)
- **screening** (*Callable[[object, float, float, float | None], float] | None*)
- **weak_rates** (*Sequence[object] | None*)

nucnetpy.solver.**jacobian** (*network*, *species*, *thermo*, *screening=None*, *weak_rates=None*, *eps=1e-08*, *sparse=False*)

Parameters
- **network** (*Network*)
- **species** (*Sequence[str]*)
- **thermo** (*Callable[[float, Mapping[str, float]], Tuple[float, float]]*)
- **screening** (*Callable[[object, float, float, float | None], float] | None*)
- **weak_rates** (*Sequence[object] | None*)
- **eps** (*float*)
- **sparse** (*bool*)

nucnetpy.solver.**analytic_jacobian** (*network*, *species*, *thermo*, *screening=None*, *weak_rates=None*, *sparse=True*)

Return an analytic Jacobian of the stoichiometric right-hand side.

Differentiating the flow of Eq. (flow),

F_r = (lambda_r rho^(n_r-1) / prod_i m_ir!) prod_i Y_i^{m_ir},

with respect to a reactant abundance gives

dF_r/dY_j = m_jr Y_j^{m_jr-1} (lambda_r rho^(n_r-1) / prod_i m_ir!)
prod_{i != j} Y_i^{m_ir},

so that `J_ij = sum_r nu_ir dF_r/dY_j`. The partial product is formed explicitly rather than as `F_r m_jr / Y_j` so that the derivative stays correct when `Y_j` vanishes, which is the usual state of most species at the start of a calculation.

The whole matrix costs one sweep over the reaction list, whereas the finite-difference Jacobian of `jacobian()` costs `N+1` evaluations of the full right-hand side. The saving therefore grows linearly with the number of evolved species.

`lambda_r` is treated as independent of composition. With a screening function or weak-rate tables that depend on `Ye` the result is an approximate Jacobian: it changes the Newton convergence rate of an implicit solver but not the solution it converges to.

Parameters

- **network** (*Network*)
- **species** (*Sequence[str]*)
- **thermo** (*Callable[[float, Mapping[str, float]], Tuple[float, float]]*)
- **screening** (*Callable[[object, float, float, float | None], float] | None*)
- **weak_rates** (*Sequence[object] | None*)
- **sparse** (*bool*)

nucnetpy.solver.**jacobian_sparsity** (*network*, *species*)

Parameters

- **network** (*Network*)
- **species** (*Sequence[str]*)

nucnetpy.solver.**evolve_zone** (*network*, *zone*, *times*, *thermo=None*, *method='bdf'*, *species=None*, *screening=None*, *weak_rates=None*, *rtol=1e-06*, *atol=1e-30*, *use_jacobian=True*, *project_positive=True*, *jac_mode='analytic'*, *max_change=0.005*, *min_mass_fraction=1e-15*)

Evolve one zone over `times`.

`method="adaptive_implicit_euler"` takes implicit (backward) Euler steps solved by Newton-Raphson iterations, the scheme NucNet Tools uses, and chooses each step so that no abundance whose mass fraction exceeds `min_mass_fraction` changes by more than `max_change` (as a fraction) in one step. It is first-order accurate, so `max_change` sets the accuracy: halving it halves the error. It is the robust choice for large networks in which a fuel runs out. There the higher-order BDF formula can predict a slightly negative abundance of the fuel; the right-hand side treats it as zero, which stops every reaction that consumes the fuel, and SciPy's step size collapses. Implicit Euler cannot turn a decaying abundance negative. `rtol`, `atol`, `use_jacobian` and `jac_mode` do not apply to it.

`jac_mode` selects how the Jacobian is supplied to the implicit SciPy solvers. `"analytic"` (the default) uses `analytic_jacobian()`, which differentiates the stoichiometric flows in closed form and costs one sweep over the reaction list. `"numerical"` uses the finite-difference `jacobian()`, costing `N+1` right-hand-side evaluations. `"sparsity"` supplies no Jacobian and instead passes the stoichiometric sparsity pattern so that SciPy estimates the matrix by grouped finite differences.

Note that SciPy applies `jac_sparsity` only when no Jacobian callable is given, so the pattern and an explicit Jacobian are alternatives rather than complements.

`method` is one of `METHODS`, in any case. An unrecognised name raises `ValueError` rather than falling back to another integrator.

Parameters

- **network** (*Network*)
- **zone** (*Zone*)
- **times** (*Sequence[float]*)
- **thermo** (*Callable[[float, Mapping[str, float]], Tuple[float, float]] | None*)
- **method** (*str*)
- **species** (*Sequence[str] | None*)

- **screening** (*Callable[[object, float, float, float | None], float] | None*)
- **weak_rates** (*Sequence[object] | None*)
- **rtol** (*float*)
- **atol** (*float*)
- **use_jacobian** (*bool*)
- **project_positive** (*bool*)
- **jac_mode** (*str*)
- **max_change** (*float*)
- **min_mass_fraction** (*float*)

Return type *EvolutionResult*

nucnetpy.solver.**evolve_network_zones** (*network*, *times*, *thermo=None*, *method='bdf'*, ***kwargs*)

Parameters
- **network** (*Network*)
- **times** (*Sequence[float]*)
- **thermo** (*Callable[[float, Mapping[str, float]], Tuple[float, float]] | None*)
- **method** (*str*)

Return type *List*[*EvolutionResult*]

6.3 Equilibrium

Nuclear statistical equilibrium (NSE) and equilibrium helpers.

This module is a pure-Python counterpart to the most commonly used libnuceq workflows. It solves for chemical potentials that reproduce a requested `rho`, `T9` and `Ye` and then returns NSE abundances for the species that are present in a `nucnetpy.Network`.

The formulas are intentionally transparent and unit-aware. They are suitable for replacement workflows and regression testing, but exact equality with a specific compiled libnuceq build should be checked with the same nuclear masses, partition functions, Coulomb corrections, and numerical tolerances.

nucnetpy.nse.**nse_prefactor** (*sp*, *t9*, *rho*, *include_partition=True*)

Return the NSE abundance prefactor excluding proton/neutron potentials.

The abundance is represented as `Y_i = prefactor_i * exp(Z_i*mu_p/kT + N_i*mu_n/kT)` where `mu_p` and `mu_n` are solved dimensionless chemical potential parameters in the internal solver. The mass excess term is included with the sign convention common in NSE abundance formulas.

Parameters
- **sp** (*Species*)
- **t9** (*float*)
- **rho** (*float*)
- **include_partition** (*bool*)

Return type float

class nucnetpy.nse.**NSEResult** (*t9: 'float', rho: 'float', ye: 'float', mu_p: 'float', mu_n: 'float', abundances: 'Dict[str, float]', success: 'bool', message: 'str' = ''*)

Bases: object

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)
- **mu_p** (*float*)
- **mu_n** (*float*)
- **abundances** (*Dict[str, float]*)
- **success** (*bool*)
- **message** (*str*)

t9: float

rho: float

ye: float

mu_p: float

mu_n: float

abundances: Dict[str, float]

success: bool

message: str = ''

zone ( )

Return type *Zone*

property xsum: float

property computed_ye: float

nucnetpy.nse.**solve_nse** (*network, t9, rho, ye, species=None, include_partition=True, tol=1e-08, max_iter=200, nse_correction=None, require_nuclear_data=True*)

Solve an NSE composition for a network.

The constraints are `sum(A_i Y_i) = 1` and `sum(Z_i Y_i) = Ye`. The solve works on well-scaled residuals with a numerically stable log-sum-exp, which keeps the iron-peak prefactors representable; see `_solve_equilibrium()` for the method. `success` is true only when the residual norm is below `tol`.

When every species in the solve has the same ratio N/Z, only the combination `Z mu_p + N mu_n` is determined. The abundances are still unique, and the reported `mu_p` and `mu_n` are the minimum-norm pair, so they do not depend on the SciPy version.

`nse_correction` is an optional callable `(species, t9, rho, ye) -> f_corr` added to each species' NSE exponent, the libnucnet NSE correction factor hook. Pass `nucnetpy.coulomb.nse_correction()` for the Bravo & Garcia-Senz Coulomb correction.

Parameters
- **network** (*Network*)
- **t9** (*float*)
- **rho** (*float*)

- **ye** (*float*)
- **species** (*Sequence[str] | None*)
- **include_partition** (*bool*)
- **tol** (*float*)
- **max_iter** (*int*)
- **require_nuclear_data** (*bool*)

Return type *NSEResult*

nucnetpy.nse.**equilibrium_ratio** (*reaction*, *network*, *t9*, *rho*, *ye*)

Return product/reactant NSE abundance ratio for a reaction.

Parameters
- **network** (*Network*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

Quasi-statistical equilibrium (QSE) with constrained clusters.

A pure-Python port of the libnuceq cluster workflow used by the NucNet Tools examples (`examples/analysis/compare_equil.cpp`) and the blog posts on cluster flows, (n,gamma)-(gamma,n) equilibrium, and comparing network calculations to equilibrium.

In libnuceq an equilibrium *cluster* is a subset of species (selected by an XPath expression such as `[z >= 6]`) whose total abundance is constrained to a prescribed value. Each cluster gets one extra chemical-potential multiplier `lambda_c` on top of the global proton/neutron potentials:

Y_i = pref_i * exp(Z_i mu_p + N_i mu_n + lambda_{c(i)})

subject to `sum A_i Y_i = 1`, `sum Z_i Y_i = Ye`, and `sum_{i in c} Y_i = Y_c` for every cluster `c`. Full NSE is the special case with no clusters; an (n,gamma)-(gamma,n) r-process equilibrium is one cluster holding all heavy nuclei with `Y_c` equal to the current heavy-nuclei abundance.

Cluster membership is by species name; use `nucnetpy.network_limiter.select_species()` for XPath-like Z/A selection.

class nucnetpy.qse.**QSECluster** (*species*, *constraint*, *label=''*)

Bases: `object`

A constrained equilibrium cluster: species names and their total Y.

Parameters
- **species** (*Sequence[str]*)
- **constraint** (*float*)
- **label** (*str*)

species: **Sequence[str]**

constraint: **float**

label: **str** = **''**

class nucnetpy.qse.**QSEResult** (*t9: 'float'*, *rho: 'float'*, *ye: 'float'*, *mu_p: 'float'*, *mu_n: 'float'*, *lambdas: 'List[float]'*, *abundances: 'Dict[str, float]'*, *success: 'bool'*, *message: 'str' = ''*)

Bases: `object`

Parameters
- **t9** (*float*)

- **rho** (*float*)
- **ye** (*float*)
- **mu_p** (*float*)
- **mu_n** (*float*)
- **lambdas** (*List[float]*)
- **abundances** (*Dict[str, float]*)
- **success** (*bool*)
- **message** (*str*)

t9: **float**

rho: **float**

ye: **float**

mu_p: **float**

mu_n: **float**

lambdas: **List**[**float**]

abundances: **Dict**[**str**, **float**]

success: **bool**

message: **str** = ''

zone ( )

Return type *Zone*

property xsum: **float**

property computed_ye: **float**

nucnetpy.qse.**solve_qse** (*network*, *t9*, *rho*, *ye*, *clusters*, *species=None*, *include_partition=True*, *nse_correction=None*, *tol=1e-08*, *max_iter=300*, *require_nuclear_data=True*)

Solve a constrained (QSE) equilibrium for a network.

`clusters` is a sequence of `QSECluster`; species not in any cluster follow unconstrained NSE. A species may belong to at most one cluster. With `clusters=[]` the result equals `solve_nse()`, and the same solver is used, so `success` means the residual norm is below `tol` and non-unique potentials are reported as the minimum-norm solution. Species without nuclear data are left out, as in `solve_nse()`, unless `require_nuclear_data` is false.

Parameters

- **network** (*Network*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)
- **clusters** (*Sequence[QSECluster]*)
- **species** (*Sequence[str] | None*)
- **include_partition** (*bool*)
- **tol** (*float*)
- **max_iter** (*int*)

- **require_nuclear_data** (*bool*)

Return type *QSEResult*

nucnetpy.qse.**cluster_abundance** (*abundances*, *cluster_species*)

Return `Y_c`, the summed abundance of the cluster members.

Parameters
- **abundances** (*Mapping[str, float]*)
- **cluster_species** (*Sequence[str]*)

Return type float

nucnetpy.qse.**cluster_ydot** (*network*, *abundances*, *cluster_species*, *t9*, *rho=1.0*, *use_reverse=False*)

Return per-reaction contributions to `dY_c/dt` for a cluster.

Each reaction contributes `flux * (net change in cluster member count)`; reactions that only shuffle abundance inside the cluster contribute nothing (blog: "Calculating cluster flows", C++ `compute_Ycdot`). With `use_reverse=True` the net (forward minus detailed-balance reverse) flux is used. `sum(result.values())` is dY_c/dt.

Parameters
- **network** (*Network*)
- **abundances** (*Mapping[str, float]*)
- **cluster_species** (*Sequence[str]*)
- **t9** (*float*)
- **rho** (*float*)
- **use_reverse** (*bool*)

Return type *Dict*[str, float]

Coulomb (plasma) corrections to NSE, ported from NucNet Tools.

This is a direct port of `user/nse_corr.cpp` (r647): the Coulomb correction of Bravo & Garcia-Senz (MNRAS 307, 984, 1999) with the fit constants of Ogata & Ichimaru (Phys. Rev. A 36, 5451, 1987) and the one-component-plasma value `f(Gamma=1)/kT = -0.420` of Slattery, Doolen & DeWitt (1980).

The species Coulomb chemical potential per kT is

> Gamma_i > 1: mu_C/kT = a G + 4b G^(1/4) - 4c G^(-1/4) + d ln G - o Gamma_i <= 1: mu_C /kT = (beta/gamma) G^gamma - G^(3/2)/sqrt(3)

with `G = Gamma_i = Z^(5/3) Gamma_e` and the two branches matched in value and first derivative at `Gamma_i = 1`. `Gamma_e = e^2 / (a_e k T)` uses the electron-cloud radius `a_e = (4 pi rho N_A Ye / 3)^(-1/3)`.

The NSE correction factor is `f_corr = -mu_C/kT` (libnucnet convention): it is added to the exponent of the NSE abundance of each species and to the equilibrium constants used for detailed-balance reverse rates.

nucnetpy.coulomb.**gamma_e** (*t9*, *rho*, *ye*)

Return the electron Coulomb coupling parameter `Gamma_e`.

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.coulomb.**species_coulomb_chemical_potential** (*z*, *t9*, *rho*, *ye*)

Return `mu_C/kT` for a species of charge `z` (0 for neutrons).

Parameters
- **z** (*int*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.coulomb.**species_coulomb_energy** (*z*, *t9*, *rho*, *ye*)
Return the species Coulomb energy per particle in units of kT.

Parameters
- **z** (*int*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.coulomb.**species_coulomb_entropy** (*z*, *t9*, *rho*, *ye*)
Return the species Coulomb entropy per particle in units of k_B.

Parameters
- **z** (*int*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.coulomb.**nse_correction** (*species*, *t9*, *rho*, *ye*)
Return the NSE correction factor `f_corr = -mu_C/kT` for a species.
Signature-compatible with the `nse_correction` hooks accepted by `nucnetpy.nse.solve_nse()` and the detailed-balance helpers: pass this function (or your own with the same signature) to enable Coulomb corrections.

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.coulomb.**coulomb_entropy_per_nucleon** (*abundances*, *species_map*, *t9*, *rho*, *ye*)
Return the total Coulomb entropy per nucleon in units of k_B.
`sum_i Y_i * s_C(Z_i)` — the plasma part of the entropy used in the blog workflow "Computing NSE corrections to the entropy".

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

6.4 Detailed balance

Reverse reaction rates from detailed balance.

This replaces the libnucnet behaviour of computing reverse rates from the forward rate, nuclear masses, and partition functions (blog workflows "Comparing forward and reverse reaction rates", "Computing reaction flows", and the (n,gamma)-(gamma,n) equilibrium studies).

For a reaction `sum_r nu_r R -> sum_p nu_p P` that conserves Z and A, the equilibrium abundance ratio follows from the same Saha prefactors used by the NSE solver: `ln K = sum_p nu_p ln(pref_p) - sum_r nu_r ln(pref_r)` where `Y_i^eq = pref_i * exp(Z_i mu_p + N_i mu_n)` and the chemical-potential terms cancel for a balanced reaction. Matching forward and reverse fluxes at equilibrium gives

    lambda_rev = lambda_fwd * rho**(n_r - n_p) * (s_rev / s_fwd) / K

with `n` the reactant/product orders and `s` the duplicate-particle statistical factors. Because `ln(pref)` contains `-ln(rho * N_A)`, the explicit density powers cancel and the reverse rate is a pure function of temperature, as it must be.

Photons and leptons (`gamma`, `electron`, …) are ignored on either side, so the reverse of a radiative capture is the photodisintegration rate.

`nucnetpy.detailed_balance.`**`apply_reverse_partition_functions`** `(` *`network`*, *`rtol=1e-06`* `)`

Multiply library reverse rates by the partition functions they omit.

ReacLib supplies a reverse rate for most reactions, flagged `v` in its own format. The library derives it from the forward rate by detailed balance using the ground-state spins only, so the temperature-dependent partition functions are left out and must be applied by the user: the reverse rate of `A -> B` is the fitted value times $\prod_B G_i(T) / \prod_A G_i(T)$. Evaluated without that factor, a network relaxes to an equilibrium in which no nuclide has excited states, which at $T_9 = 5$ differs from nuclear statistical equilibrium by up to 48 per cent in the alpha-chain mass fractions.

libnucnet files do not carry the `v` flag, so each derived reverse rate is recognised from its coefficients (see `_is_library_reverse()`) and given a `PartitionRatio` built from the partition functions in `network.species`. Species without a table count as $G = 1$. The XML readers call this function by default; call it yourself for a network assembled in memory, or pass `reverse_partition_functions=False` to a reader to keep the uncorrected rates.

Returns the list of reactions that were corrected.

Parameters **`rtol`** (*`float`*)

Return type list

`nucnetpy.detailed_balance.`**`log_equilibrium_constant`** `(` *`reaction`*, *`species_map`*, *`t9`*, *`rho=1.0`*, *`include_partition=True`* `)`

Return `ln K` with `K = prod Y_p^eq / prod Y_r^eq` for the reaction.

Requires mass excesses (and optionally partition functions) on the species; the Q-value enters through the mass excesses as `exp(Q/kT)`.

Parameters

- **`reaction`** (*`Reaction`*)
- **`species_map`** (*`Mapping[str, Species]`*)
- **`t9`** (*`float`*)
- **`rho`** (*`float`*)
- **`include_partition`** (*`bool`*)

Return type float

nucnetpy.detailed_balance.**reverse_rate** (*reaction*, *species_map*, *t9*, *rho=1.0*, *forward=None*, *include_partition=True*)

Return the detailed-balance reverse rate for `reaction` at `t9`.

`forward` overrides the forward rate (default `reaction.rate(t9)`). The result is in the same convention as forward rates: to obtain a flux it must be combined with `rho**(n_p - 1)`, product abundances, and the product-side statistical factor (see `net_flows()`).

Parameters
- **reaction** (*Reaction*)
- **species_map** (*Mapping[str, Species]*)
- **t9** (*float*)
- **rho** (*float*)
- **forward** (*float | None*)
- **include_partition** (*bool*)

Return type float

nucnetpy.detailed_balance.**reverse_reaction** (*reaction*, *species_map*, *t9_grid=None*, *include_partition=True*)

Return a Reaction for the reverse process with a tabulated rate.

The rate is sampled from `reverse_rate()` on `t9_grid` (default 30 points, T9 = 0.1 .. 10) so the reverse process — e.g. the (gamma,n) partner of an (n,gamma) capture — can be added to a network like any other reaction.

Parameters
- **reaction** (*Reaction*)
- **species_map** (*Mapping[str, Species]*)
- **t9_grid** (*Sequence[float] | None*)
- **include_partition** (*bool*)

Return type *Reaction*

nucnetpy.detailed_balance.**consistent_reverse_network** (*network*, *t9_grid=None*, *include_partition=True*, *tabulate=False*)

Return a copy of `network` whose reverse rates obey detailed balance.

Rate libraries such as JINA ReacLib derive each reverse rate from its forward rate by detailed balance, but with the library's own Q value, with ground-state spins only, and as a fitted expression. Once the partition functions are restored (`apply_reverse_partition_functions()`, applied by the XML readers), the library reverse rates of the alpha chain at $T_9 = 5$ agree with the detailed-balance value implied by the nuclide file to about one per cent, which still leaves the stationary state of the network slightly away from the nuclear statistical equilibrium of those same data.

This function pairs every reaction with its inverse, keeps the exothermic member of each pair, judged from the nuclide masses because libnucnet files carry no Q value, and regenerates the endothermic member from `reverse_rate()`. Reactions with no inverse in the network are carried over unchanged. The resulting network relaxes to the composition returned by `nucnetpy.solve_nse()` to about one part in 10^7, so the integrator and the equilibrium solver become mutually consistent.

By default the reverse rate is attached as a callable evaluated exactly at whatever temperature the integrator asks for. This matters: the reverse rate carries a factor $\exp(-Q/kT)$, which is far too steep to interpolate accurately, and tabulating it on 200 logarithmically spaced points degrades the agreement with equilibrium from about one part in 10^7 to about one part in 10^3. Pass `tabulate=True` to sample the rate on `t9_grid` instead, which is less accurate but can be written to XML.

Note that this replaces the library's reverse rates, which carry the library's Q values and the error of its fits, with values derived from the masses in the nuclide file, so it enforces thermodynamic

consistency with that file rather than improving individual rates.

Parameters
- **t9_grid** (*Sequence[float] | None*)
- **include_partition** (*bool*)
- **tabulate** (*bool*)

nucnetpy.detailed_balance.**reverse_flux** (*reaction*, *species_map*, *abundances*, *t9*, *rho=1.0*, *include_partition=True*)

Return the detailed-balance reverse flux of `reaction` at this composition.
This is the flux the inverse reaction would carry with the rate from `reverse_rate()`, in the same units as `Reaction.flux()`.

Parameters
- **reaction** (*Reaction*)
- **species_map** (*Mapping[str, Species]*)
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)
- **include_partition** (*bool*)

Return type float

nucnetpy.detailed_balance.**net_flows** (*network*, *abundances*, *t9*, *rho=1.0*, *include_partition=True*)

Return `{reaction: (forward, reverse, net)}` fluxes for a network.
The forward flux follows `Reaction.flux()`; the reverse flux uses the detailed-balance rate with the product abundances. At NSE abundances the net flux of every balanced reaction vanishes.
Every listed reaction is given its own detailed-balance reverse, including one whose inverse the network also lists explicitly, as JINA networks do. The entries for such a pair therefore describe the same process twice; do not add them up. `nucnetpy.analysis.entropy_generation_rate()` and `nucnetpy.analysis.integrated_currents()` handle pairs correctly.

Parameters
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)
- **include_partition** (*bool*)

Return type *Dict*[str, *Tuple*[float, float, float]]

6.5 Screening, weak rates, decays

Electron-screening corrections for thermonuclear reaction rates.

The original NucNet Tools exposes user-defined screening callbacks. This module provides a Python callback-compatible implementation with common weak and intermediate-screening approximations. The API returns multiplicative factors applied to bare nuclear rates.

nucnetpy.screening.**electron_number_density** (*rho*, *ye*)

Parameters
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.screening.**debye_radius** (*t9*, *rho*, *ye*, *ion_strength=0.0*)

Return Debye radius in cm.

`ion_strength` may be supplied as `sum Z_i^2 Y_i`. If omitted, the electron term alone is used, which is the usual safe weak-screening fallback.

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)
- **ion_strength** (*float*)

Return type float

nucnetpy.screening.**ion_strength** (*abundances*, *species_map=None*)

Parameters
- **abundances** (*Mapping[str, float]*)
- **species_map** (*Mapping[str, Species] | None*)

Return type float

class nucnetpy.screening.**ScreeningContext** (*t9: 'float', rho: 'float', ye: 'float', abundances: 'Mapping[str, float]', species_map: 'Optional[Mapping[str, Species]]' = None*)

Bases: `object`

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)
- **abundances** (*Mapping[str, float]*)
- **species_map** (*Mapping[str, Species] | None*)

t9: **float**

rho: **float**

ye: **float**

abundances: **Mapping**[**str**, **float**]

species_map: **Mapping**[**str**, **Species**] | **None** = **None**

property ion_strength: **float**

nucnetpy.screening.**weak_screening_factor** (*z1*, *z2*, *t9*, *rho*, *ye*, *ion_strength_value=0.0*)

Salpeter weak-screening enhancement factor.

The factor is `exp(Z1 Z2 e^2 / (kT R_D))`. A cap prevents numerical overflow while retaining monotonic enhancement.

Parameters
- **z1** (*int*)
- **z2** (*int*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

- **ion_strength_value** (*float*)

Return type float

nucnetpy.screening.**graboske_intermediate_factor** (*z1*, *z2*, *t9*, *rho*, *ye*, *ion_strength_value=0.0*)

Smooth weak-to-intermediate approximation inspired by Graboske et al.

This is not tied to one compiled NucNet user file; it gives a stable, callback-compatible enhancement and can be replaced by a user function when exact project-specific screening is required.

Parameters

- **z1** (*int*)
- **z2** (*int*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)
- **ion_strength_value** (*float*)

Return type float

nucnetpy.screening.**reaction_screening_factor** (*reaction*, *context*, *model='weak'*)

Parameters

- **context** (*ScreeningContext*)
- **model** (*str*)

Return type float

class nucnetpy.screening.**SkyNetScreening** (*species_map=None*, *max_z=110*, *pair_term=None*)

Bases: `object`

Screening as a Coulomb chemical potential per nuclear charge.

Instead of a pairwise enhancement factor, each charge `Z` is assigned a Coulomb chemical potential `mu(Z)/kT` in the plasma, and a reaction is enhanced by

exp(sum_reactants mu(Z_i)/kT - mu(sum_i Z_i)/kT),

the difference between the potentials of the separated reactants and of the fused compound charge. This has two advantages over a pairwise factor: it extends naturally to reactions with more than two charged reactants, and because it is a chemical potential it can be applied to the forward and reverse directions consistently, so screening does not break detailed balance.

`mu(Z)` blends three regimes with smooth `tanh` switches: weak screening, `mu_w = -Z^2 zeta lambda_0 / 2`; intermediate screening, `mu_i = -0.380 lambda_0^b eta_b Z^(b+1)` with `b = 0.860`; and strong screening from a one-component-plasma fit.

`zeta` is the Debye sum per particle, `zeta^2 = (sum Z^2 Y + E) / sum Y`, where `E` is the electrons' and positrons' share. As in SkyNet, `E` comes from the electron gas: by default (`pair_term=None`) it is computed at every update by `nucnetpy.electrons` as `kT (dn /dmu) / (rho N_A)`. That is `Ye` for non-degenerate electrons, which recovers Salpeter's `zeta^2 = sum (Z^2 + Z) Y`; it falls towards zero as the electrons become degenerate and no longer screen, and it rises above `Ye` when positron pairs add to the screening. A number passed as `pair_term` is used instead, so `pair_term=0.0` gives the ions-only (fully degenerate) limit.

The electron state is cached and only recomputed when the temperature, density or `Ye` change (`Ye` is compared to 10 decimal places).

The object is a callable suitable for the `screening` argument of `nucnetpy.evolve_zone()`. It caches `mu(Z)` and recomputes it in `update()`, which the network calls once per right-hand-side evaluation, so the cost does not scale with the number of reactions.

Parameters

- **species_map** (*Optional[Mapping[str, Species]]*)

- **max_z** (*int*)
- **pair_term** (*Optional[float]*)

electron_term (*t9*, *rho*, *ye*)

Return E, the electrons' and positrons' share of the Debye sum.
This is pair_term if one was given, and otherwise nucnetpy.electrons.ElectronGas.screening_term at (t9, rho, ye).

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

update (*abundances*, *t9*, *rho*, *ye=None*)

Recompute mu(Z) for this composition and thermodynamic state.

Parameters
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float | None*)

Return type None

chemical_potential (*z*)

Return mu(Z)/kT for charge z.

Parameters **z** (*int*)

Return type float

factor (*reaction*)

Return the multiplicative screening enhancement of reaction.

Return type float

Weak-rate and neutrino helper layer.

Supports two-dimensional T9/rhoYe tables, simple FFN/Oda/Langanke-style text blocks after normalization, bilinear interpolation, and ODE source-term hooks.

class nucnetpy.weak.**WeakRateTable** (*parent: 'str', daughter: 'str', t9: 'Sequence[float]', rho_ye: 'Sequence[float]', rates: 'np.ndarray', nu_loss: 'Optional[np.ndarray]' = None, label: 'str' = ''*)

Bases: object

Parameters
- **parent** (*str*)
- **daughter** (*str*)
- **t9** (*Sequence[float]*)
- **rho_ye** (*Sequence[float]*)
- **rates** (*ndarray*)
- **nu_loss** (*ndarray | None*)
- **label** (*str*)

parent: **str**

daughter: str

t9: Sequence[float]

rho_ye: Sequence[float]

rates: ndarray

nu_loss: ndarray | None = None

label: str = ''

rate (*t9*, *rho_ye*)

Parameters
- **t9** (*float*)
- **rho_ye** (*float*)

Return type float

neutrino_loss (*t9*, *rho_ye*)

Parameters
- **t9** (*float*)
- **rho_ye** (*float*)

Return type float

nucnetpy.weak.**read_weak_table** (*path*, *parent=None*, *daughter=None*)

Read a normalised weak-rate table.
Accepted format:

```
# parent fe56 daughter mn56
t9 rhoYe rate [nu_loss]
1.0 1e7 2.3e-4 1.0e-5
```

Repeated rows are reshaped into a rectangular T9 x rhoYe grid.

Parameters
- **path** (*str | Path*)
- **parent** (*str | None*)
- **daughter** (*str | None*)

Return type *WeakRateTable*

nucnetpy.weak.**read_weak_tables** (*paths*)

Parameters **paths** (*Iterable[str | Path]*)

Return type *List*[*WeakRateTable*]

nucnetpy.weak.**compute_yedot** (*abundances*, *weak_rates*, *species_map*, *t9*, *rho*, *ye*)

Parameters
- **abundances** (*Mapping[str, float]*)
- **weak_rates** (*Sequence[WeakRateTable]*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.weak.**weak_source_terms** (*abundances*, *weak_rates*, *t9*, *rho*, *ye*)

Parameters
- **abundances** (*Mapping[str, float]*)
- **weak_rates** (*Sequence[WeakRateTable]*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type *Dict*[str, float]

Nuclear decay helpers ported into Python.

class nucnetpy.decay.**DecayRecord** (*parent: 'str'*, *daughter: 'str'*, *half_life: 'float'*, *branch: 'float' = 1.0*, *mode: 'str' = 'decay'*)

Bases: `object`

Parameters
- **parent** (*str*)
- **daughter** (*str*)
- **half_life** (*float*)
- **branch** (*float*)
- **mode** (*str*)

parent: **str**

daughter: **str**

half_life: **float**

branch: **float** = **1.0**

mode: **str** = **'decay'**

property rate: **float**

reaction ( )

Return type *Reaction*

nucnetpy.decay.**add_decay_records** (*network*, *records*)

Parameters **records** (*Iterable[DecayRecord]*)

Return type None

nucnetpy.decay.**decay_constant_from_half_life** (*half_life*)

Parameters **half_life** (*float*)

Return type float

nucnetpy.decay.**fission_reaction** (*parent*, *fragments*, *neutrons=0*, *rate=0.0*, *mode='sf'*)

Return a fission reaction `parent -> fragments + neutrons * n`.
`rate` is the fission decay constant in 1/s (use `decay_constant_from_half_life()` for spontaneous-fission half-lives). `mode` labels the channel, e.g. `"sf"` (spontaneous) or `"bdf"` (beta-delayed); for neutron-induced fission build a normal two-reactant Reaction instead. Blog workflow: “Adding fission to an r-process calculation”.

Parameters
- **parent** (*str*)

- **fragments** (*Iterable[str]*)
- **neutrons** (*int*)
- **rate** (*float*)
- **mode** (*str*)

Return type *Reaction*

Neutrino reaction-rate helper functions.

class nucnetpy.neutrino.**NeutrinoLuminosity** (*l0: 'float', tau: 'float | None' = None*)

Bases: object

Parameters
- **l0** (*float*)
- **tau** (*float | None*)

l0: float

tau: float | None = None

luminosity (*t*)

Parameters **t** (*float*)

Return type float

class nucnetpy.neutrino.**NeutrinoQuantity** (*temperature: 'Sequence[float]', log10_xsec: 'Sequence[float]'*)

Bases: object

Parameters
- **temperature** (*Sequence[float]*)
- **log10_xsec** (*Sequence[float]*)

temperature: Sequence[float]

log10_xsec: Sequence[float]

cross_section (*tnu*)

Parameters **tnu** (*float*)

Return type float

nucnetpy.neutrino.**geometric_flux_rate** (*luminosity, mean_energy_mev, radius_cm, cross_section_cm2*)

Parameters
- **luminosity** (*float*)
- **mean_energy_mev** (*float*)
- **radius_cm** (*float*)
- **cross_section_cm2** (*float*)

Return type float

6.6 Analysis

Analysis helpers equivalent to common NucNet Tools example programs.

nucnetpy.analysis.**largest_mass_fractions** (*zone*, *species_map*, *n=10*, *min_x=0.0*)

Parameters
- **zone** (*Zone*)
- **species_map** (*Mapping[str, Species]*)
- **n** (*int*)
- **min_x** (*float*)

nucnetpy.analysis.**abundances_for_element** (*zone*, *element*, *species_map*)

Parameters
- **zone** (*Zone*)
- **element** (*str*)
- **species_map** (*Mapping[str, Species]*)

Return type *Dict*[str, float]

nucnetpy.analysis.**abundances_vs_nucleon_number** (*zone*, *species_map*)

Parameters
- **zone** (*Zone*)
- **species_map** (*Mapping[str, Species]*)

Return type *Dict*[int, float]

nucnetpy.analysis.**element_abundances** (*zone*, *species_map*)

Parameters
- **zone** (*Zone*)
- **species_map** (*Mapping[str, Species]*)

Return type *Dict*[str, float]

nucnetpy.analysis.**abundance_moment** (*zone*, *species_map*, *moment=1*, *kind='a'*)

Parameters
- **zone** (*Zone*)
- **species_map** (*Mapping[str, Species]*)
- **moment** (*int*)
- **kind** (*str*)

Return type float

nucnetpy.analysis.**species_history** (*network*, *species*)

Parameters
- **network** (*Network*)
- **species** (*str*)

Return type *List*[*Tuple*[int, *Tuple*[str, str, str], float]]

nucnetpy.analysis.**flows** (*network*, *zone_index=0*, *t9=None*, *rho=None*)

Parameters
- **network** (*Network*)
- **zone_index** (*int*)
- **t9** (*float | None*)
- **rho** (*float | None*)

nucnetpy.analysis.**ydot** (*network*, *zone_index=0*, *t9=None*, *rho=None*)

Parameters
- **network** (*Network*)
- **zone_index** (*int*)
- **t9** (*float | None*)
- **rho** (*float | None*)

nucnetpy.analysis.**energy_generation_rate** (*network*, *zone_index=0*, *t9=None*, *rho=None*)

Parameters
- **network** (*Network*)
- **zone_index** (*int*)
- **t9** (*float | None*)
- **rho** (*float | None*)

Return type float

nucnetpy.analysis.**compare_rates** (*net_a*, *net_b*, *t9*, *rho=1.0*)

Parameters
- **net_a** (*Network*)
- **net_b** (*Network*)
- **t9** (*float*)
- **rho** (*float*)

Return type *List*[*Tuple*[str, float, float, float]]

nucnetpy.analysis.**separation_energy** (*species_name*, *species_map*, *particle='n'*)

Return S_n or S_p in MeV, or None if the needed nuclides are absent.
`S_n(Z, A) = ME(Z, A-1) + ME(n) - ME(Z, A)` and `S_p(Z, A) = ME(Z-1, A-1) + ME(p) - ME(Z, A)`.

Parameters
- **species_name** (*str*)
- **species_map** (*Mapping[str, Species]*)
- **particle** (*str*)

Return type float | None

nucnetpy.analysis.**charge_changing_flows** (*network*, *zone_index=0*, *t9=None*, *rho=None*)

Return each reaction's contribution to dYe/dt (flux times net ΔZ).
Only reactions with a nonzero nuclear charge change (weak reactions such as beta decays and electron captures) appear in the result; strong reactions conserve Z and contribute nothing. `sum(result.values())` is dYe/dt.

Parameters
- **network** (*Network*)
- **zone_index** (*int*)
- **t9** (*float | None*)
- **rho** (*float | None*)

Return type *Dict*[str, float]

nucnetpy.analysis.**system_timescales** (*network*, *zone_index=0*, *t9=None*, *rho=None*)

Return per-species timescales `Y / |dY/dt|` in seconds.

Species with zero derivative get `inf`. The shortest timescales identify the stiffest components of the system (blog: "Computing system timescales").

Parameters
- **network** (*Network*)
- **zone_index** (*int*)
- **t9** (*float | None*)
- **rho** (*float | None*)

Return type *Dict*[str, float]

nucnetpy.analysis.**heavy_nuclei_abundance** (*zone*, *species_map*, *zmin=3*)

Return Y_h, the total abundance of nuclei with Z >= `zmin`.

The photon-to-heavy-nucleus ratio and the r-process neutron-to-seed ratio are built from this quantity (blog: "Computing the number of heavy nuclei").

Parameters
- **zone** (*Zone*)
- **species_map** (*Mapping[str, Species]*)
- **zmin** (*int*)

Return type float

nucnetpy.analysis.**neutron_exposure** (*result*, *thermo*)

Return the s-process neutron exposure tau = ∫ n_n v_T dt in mb^-1.

`result` is an `EvolutionResult` whose species include the neutron; `thermo` is the same `(t, abundances) -> (t9, rho)` function used for the evolution. `n_n = rho N_A Y_n` and the thermal velocity is `v_T = sqrt(2 k T / m_n)` (blog: "Computing the s-process neutron exposure").

Return type float

nucnetpy.analysis.**reaction_entropy_changes** (*network*, *abundances*, *t9*, *rho*)

Return per-reaction entropy change ΔS in units of k_B per reaction.

`ΔS = -sum_i nu_i mu_i/kT` with Maxwell–Boltzmann chemical potentials `mu_i/kT = ln Y_i - ln pref_i`; this equals the C++ per-reaction `Q/kT + sum_r ln(Y/Y_Q) - sum_p ln(Y/Y_Q)` of NucNet Tools `flow_utilities.cpp` and, under detailed balance, `ln(f/r)`.

ΔS is formally infinite for a reaction that involves an absent species, because `ln Y` diverges as `Y` tends to zero. Such an abundance is replaced by 1e-300 here, so the value returned for that reaction reflects the floor rather than the physics; `entropy_generation_rate()` leaves those reactions out.

Parameters
- **network** (*Network*)
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)

Return type *Dict*[str, float]

nucnetpy.analysis.**entropy_generation_rate** (*network*, *zone_index=0*, *t9=None*, *rho=None*, *use_reverse=True*)

Return dS/dt per nucleon in units of k_B per second.

Ports NucNet Tools `compute_entropy_generation_rate`: the sum over reactions of `(f -`

r) * ΔS, where ΔS is the per-reaction entropy change of reaction_entropy_changes() (blog series "Computing the entropy generation rate"). Each forward/reverse pair is counted once. When the network lists the inverse of a reaction, as JINA networks do, r is that inverse's own flux, so the result is exactly −sum_i (mu_i/kT) dY_i/dt for the network being evolved. Otherwise r is the detailed-balance reverse flux, as in NucNet Tools. With reverse rates that obey detailed balance every term equals (f − r) ln(f/r), so the total is non-negative and vanishes at NSE.

With use_reverse=False only the listed reactions are used, each with its forward flux: sum_r f_r ΔS_r = −sum_i (mu_i/kT) dY_i/dt for exactly the network as given, with no detailed-balance reverses implied.

Reactions that involve an absent species (Y <= 1e−100, the NucNet Tools threshold) are left out. Creating a species from zero abundance produces formally infinite entropy, so any finite value for such a reaction would be set by an arbitrary floor rather than by the physics; treat the total for a composition with absent species as a lower bound. (NucNet Tools skips a reaction when its forward flux vanishes, and drops the logarithm of an absent product, which can give a term of the wrong sign.)

Electron and neutrino chemical-potential terms of the C++ version are not included; nucnetpy networks carry weak reactions with their own tabulated rates instead.

Parameters
- **network** (*Network*)
- **zone_index** (*int*)
- **t9** (*float | None*)
- **rho** (*float | None*)
- **use_reverse** (*bool*)

Return type float

nucnetpy.analysis.**integrated_currents** (*network*, *result*, *thermo*, *use_reverse=True*)

Return per-reaction time-integrated net currents over an evolution.

Ports NucNet Tools update_flow_currents: for each reaction the current accumulates (forward − reverse) * dt across the time grid of result (an EvolutionResult), integrated here with the trapezoidal rule. The integrated current of a reaction is the net number of transitions per nucleon it produced over the calculation (blog: "Creating integrated currents diagrams", "Analyzing integrated currents quantitatively").

The reverse flux is the flux of the inverse reaction when the network lists it, and the detailed-balance value otherwise. A pair listed in both directions therefore appears twice, with equal and opposite currents. With use_reverse=False each entry is the reaction's own forward current. Reactions listed more than once are added together.

Parameters
- **network** (*Network*)
- **use_reverse** (*bool*)

Return type *Dict*[str, float]

nucnetpy.analysis.**nuclear_energy_generation_rate** (*network*, *zone_index=0*, *t9=None*, *rho=None*, *screening=None*, *abundances=None*, *ydot_values=None*)

Return the nuclear energy generation rate in erg per gram per second.

The rate follows from the change in total mass excess of the composition,

eps = -N_A * C * sum_i (dY_i/dt) * Delta M_i,

with Delta M_i the mass excess in MeV and C = 1.602176634e−6 erg/MeV (constants.MEV_TO_ERG) converting the sum, which carries units of MeV/g/s, into erg/g/s. This is exact, and it is preferable to summing Q_r times the flow of each reaction: it does not use the reaction Q-values stored in the rate library, so it is internally consistent with the adopted mass-excess table – the same table the equilibrium solver uses – and energy release and equilibrium are therefore built from one set of nuclear data. It may differ from library Q-values when

the two data sources use different nuclear masses.
A positive value denotes energy released to the plasma. Neutrino losses are not subtracted: energy carried away by neutrinos from weak reactions must be accounted for separately.
`abundances` and `ydot_values` override the zone composition and its derivative, which is useful when the rate is wanted along a trajectory that has already been integrated.

Parameters
- **network** (*Network*)
- **zone_index** (*int*)
- **t9** (*float | None*)
- **rho** (*float | None*)
- **abundances** (*Mapping[str, float] | None*)
- **ydot_values** (*Mapping[str, float] | None*)

Return type float

nucnetpy.analysis.**nuclear_energy_release** (*network*, *initial*, *final*)

Return the energy released between two compositions, in erg per gram.
This is the time integral of `nuclear_energy_generation_rate()` and depends only on the endpoints, being the difference in total mass excess.

Parameters
- **network** (*Network*)
- **initial** (*Mapping[str, float]*)
- **final** (*Mapping[str, float]*)

Return type float

6.7 Input and output

Fast, permissive NucNet/JINA/libnucnet XML reader and writer.

This module is tuned for real JINA XML databases, where files may contain thousands of nuclides and tens of thousands of reactions. It supports:

- separate `<nuclear_data>` and `<reaction_data>` files,
- combined `<libnucnet_input>` output files,
- zone files with mass fractions or abundances,
- ReacLib/JINA `non_smoker_fit` blocks with `a1`–`a7` coefficients,
- single-rate beta-decay style records,
- partition-function tables.

nucnetpy.io.xml.**read_xml** (*path*, *reverse_partition_functions=True*)

Read a libnucnet-style XML file holding nuclides, reactions and zones.
With `reverse_partition_functions` (the default) the library reverse rates are multiplied by the partition functions they are fitted without; see `nucnetpy.apply_reverse_partition_functions()`.

Parameters
- **path** (*str | Path*)
- **reverse_partition_functions** (*bool*)

Return type *Network*

nucnetpy.io.xml.**read_network_xml** (*path*, *reverse_partition_functions=True*)

Read a libnucnet-style XML file holding nuclides, reactions and zones.

With `reverse_partition_functions` (the default) the library reverse rates are multiplied by the partition functions they are fitted without; see `nucnetpy.apply_reverse_partition_functions()`.

Parameters
- **path** (*str | Path*)
- **reverse_partition_functions** (*bool*)

Return type *Network*

nucnetpy.io.xml.**read_xml_root** (*root*, *source='<xml-root>'*, *reverse_partition_functions=True*)

Parameters
- **source** (*str*)
- **reverse_partition_functions** (*bool*)

Return type *Network*

nucnetpy.io.xml.**write_xml** (*network*, *path*)

Parameters
- **network** (*Network*)
- **path** (*str | Path*)

Return type None

nucnetpy.io.xml.**write_zone_xml** (*network*, *path*)

Parameters
- **network** (*Network*)
- **path** (*str | Path*)

Return type None

nucnetpy.io.xml.**read_xml_string** (*text*, *reverse_partition_functions=True*)

Parameters
- **text** (*str*)
- **reverse_partition_functions** (*bool*)

Return type *Network*

JINA/libnucnet XML database helpers.

The original NucNet Tools workflow usually reads nuclide data and reaction data from JINA ReacLib /libnucnet XML files. These helpers keep that workflow in the pure-Python port: one file may contain nuclides, another may contain reactions, and both can be merged into one `nucnetpy.core.Network` object.

nucnetpy.io.jina.**read_jina_xml** (*nuclides_xml*, *reactions_xml*, *zones_xml=None*, *reverse_partition_functions=True*)

Read separate JINA/libnucnet nuclide and reaction XML files.

Parameters
- **nuclides_xml** (*str | Path*) – XML file containing `<nuclide>` records with Z, A, mass excess, spin, and optional partition-function data.
- **reactions_xml** (*str | Path*) – XML file containing `<reaction>` records and ReacLib rate fits.
- **zones_xml** (*str | Path | None*) – Optional XML file containing zones /abundances. If omitted, the returned network contains species and reactions

but no zones.

- **reverse_partition_functions** (*bool*) – Multiply the library reverse rates by the partition functions that ReacLib fits them without (default). Pass `False` to keep the rates exactly as fitted; see `nucnetpy.apply_reverse_partition_functions()`.

Returns A single Python network object with species, reactions, and optionally zones.

Return type Network

`nucnetpy.io.jina.`**`combine_jina_xml`** (*nuclides_xml*, *reactions_xml*, *output_xml*, *zones_xml=None*)

Read separate JINA XML files and write one combined nucnetpy XML file.

Parameters

- **nuclides_xml** (*str | Path*)
- **reactions_xml** (*str | Path*)
- **output_xml** (*str | Path*)
- **zones_xml** (*str | Path | None*)

Return type *Network*

`nucnetpy.io.jina.`**`jina_database_summary`** (*net_or_nuclides*, *reactions_xml=None*, *zones_xml=None*)

Return a compact summary of a loaded JINA database.

Accepts either an already loaded Network, or file paths `(nuclides_xml, reactions_xml[, zones_xml])`.

Return type dict[str, int]

6.8 Utilities

Network limiting utilities for reduced-network calculations.

`nucnetpy.network_limiter.`**`select_species`** (*network*, *zmin=None*, *zmax=None*, *amin=None*, *amax=None*, *elements=None*)

Return species names matching Z/A ranges and/or element symbols.

A pure-Python replacement for the libnucnet nuclide XPath selections used in the blog workflows "Selecting an input network" and "Applying a nuclide XPath expression". Combine with `limit_network()`:

```
limit_network(net, select_species(net, zmin=1, zmax=28, amax=60))
```

Parameters

- **zmin** (*int | None*)
- **zmax** (*int | None*)
- **amin** (*int | None*)
- **amax** (*int | None*)
- **elements** (*Iterable[str] | None*)

Return type *List*[str]

`nucnetpy.network_limiter.`**`limit_network`** (*network*, *species*, *include_linked=False*)

Restrict `network` in place to the given nuclides and the reactions among them.

A reaction survives if every nuclide it involves is kept. Photons and leptons are not nuclides, so they never decide the outcome: a photodisintegration among kept nuclides survives whether or

not `"gamma"` is listed, and listing it is harmless. With `include_linked` the kept set grows to every nuclide that shares a reaction with it; the photon does not count as a link, or every photon reaction would join the network.

Parameters
- **species** (*Iterable[str]*)
- **include_linked** (*bool*)

Hydrodynamic trajectory utilities replacing user/hydro* helpers.

class nucnetpy.hydro.**Trajectory** (*time: 'np.ndarray', t9: 'np.ndarray', rho: 'np.ndarray'*)

Bases: object

Parameters
- **time** (*ndarray*)
- **t9** (*ndarray*)
- **rho** (*ndarray*)

time: ndarray

t9: ndarray

rho: ndarray

thermo (*t, y=None*)

Parameters **t** (*float*)

Return type *Tuple*[float, float]

classmethod from_columns (*time, t9, rho*)

Parameters
- **time** (*Sequence[float]*)
- **t9** (*Sequence[float]*)
- **rho** (*Sequence[float]*)

Return type *Trajectory*

nucnetpy.hydro.**read_trajectory** (*path*)

Parameters **path** (*str | Path*)

Return type *Trajectory*

nucnetpy.hydro.**exponential_expansion** (*t0, t9_0, rho_0, tau, n=200, t_end=None*)

Parameters
- **t0** (*float*)
- **t9_0** (*float*)
- **rho_0** (*float*)
- **tau** (*float*)
- **n** (*int*)
- **t_end** (*float | None*)

Return type *Trajectory*

Thermodynamics of the burning matter: entropy, and finding T or rho from it.

`entropy_per_nucleon()` adds up the entropy of the three parts of the matter, each treated in full:

- **ions** - the Sackur-Tetrode entropy of each nuclear species, with its real mass, ground-state spin and temperature-dependent partition function;
- **electrons and positrons** - from `nucnetpy.electrons`, at any degeneracy and including pairs;
- **photons** - black-body radiation, `4 a T^3 / (3 rho N_A k)`;

and, optionally, the Coulomb (plasma) correction of `nucnetpy.coulomb`.

`density_for_entropy()` and `t9_for_entropy()` search for the density or temperature that gives a target entropy at fixed composition, and `constant_entropy_thermo()` uses the second of these to drive a network calculation along a constant-entropy (adiabatic) expansion.

All entropies are per nucleon in units of Boltzmann's constant.

nucnetpy.thermo.**ideal_gas_pressure** (*rho*, *temperature*, *ytot*)

Parameters
- **rho** (*float*)
- **temperature** (*float*)
- **ytot** (*float*)

Return type float

nucnetpy.thermo.**radiation_pressure** (*temperature*)

Parameters **temperature** (*float*)

Return type float

class nucnetpy.thermo.**EntropyPerNucleon** (*ions*, *electrons*, *photons*, *coulomb=0.0*)

Bases: `object`

The entropy per nucleon, in units of k_B, split into its parts.

Parameters
- **ions** (*float*)
- **electrons** (*float*)
- **photons** (*float*)
- **coulomb** (*float*)

ions: float

electrons: float

photons: float

coulomb: float = 0.0

property total: float

nucnetpy.thermo.**photon_entropy_per_nucleon** (*t9*, *rho*)

Entropy of black-body radiation per nucleon, `4 a T^3 / (3 rho N_A k)`.

Parameters
- **t9** (*float*)
- **rho** (*float*)

Return type float

nucnetpy.thermo.**ion_entropy_per_nucleon** (*abundances*, *species*, *t9*, *rho*, *include_partition=True*)

Sackur-Tetrode entropy of the nuclei per nucleon, in units of k_B.

sum_i Y_i (5/2 + ln(Y_Q,i / Y_i) + d ln G_i / d ln T), where Y_Q,i = (2J_i+1) G_i(T) n_Q,i / (rho N_A) uses the nuclear mass through the quantum concentration n_Q,i = (A_i m_u kT / 2 pi hbar^2)^(3/2). The last term is the entropy of the nuclei's excited states. Species with zero abundance and massless particles contribute nothing.

Parameters
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)
- **include_partition** (*bool*)

Return type float

nucnetpy.thermo.**entropy_per_nucleon** (*abundances*, *species*, *t9*, *rho*, *ye=None*, *include_partition=True*, *electrons=True*, *photons=True*, *coulomb=False*)

Return the entropy per nucleon of the matter, split into its parts.

abundances maps species names to abundances Y; species is a Network or a mapping of names to Species, which supplies the spins and partition functions (species missing from it are parsed from their names and get a weight of one). ye defaults to sum Z Y of the abundances. t9 is the temperature in 10^9 K and rho the density in g cm^-3.

The flags choose which parts are computed; parts that are switched off are reported as zero. The Coulomb correction is off by default, as in the NSE solver.

Example: pure helium-4 at T9 = 5 and rho = 1e8 has ion entropy 3.24, electron entropy 1.07 and photon entropy 0.15 per nucleon.

Parameters
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float | None*)
- **include_partition** (*bool*)
- **electrons** (*bool*)
- **photons** (*bool*)
- **coulomb** (*bool*)

Return type *EntropyPerNucleon*

nucnetpy.thermo.**density_for_entropy** (*entropy*, *abundances*, *species*, *t9*, *ye=None*, *bounds=(1e-12, 1000000000000000.0)*, *include_partition=True*, *electrons=True*, *photons=True*, *coulomb=False*)

Return the density (g cm^-3) at which the matter has the given entropy.

The composition and the temperature t9 are held fixed, and the entropy is entropy_per_nucleon() with the same options. The entropy falls steadily as the density rises, so the answer is unique; it is found by a bracketing search in ln rho between bounds, to a relative precision of about 10^-14. A ValueError reports the entropy range those bounds can reach if the target is outside it.

Parameters
- **entropy** (*float*)
- **abundances** (*Mapping[str, float]*)
- **t9** (*float*)
- **ye** (*float | None*)
- **bounds** (*Tuple[float, float]*)
- **include_partition** (*bool*)

- **electrons** (*bool*)
- **photons** (*bool*)
- **coulomb** (*bool*)

Return type float

nucnetpy.thermo.**t9_for_entropy** (*entropy*, *abundances*, *species*, *rho*, *ye=None*, *bounds=(0.001, 100.0)*, *include_partition=True*, *electrons=True*, *photons=True*, *coulomb=False*)

Return the temperature (in 10^9 K) at which the matter has the given entropy.

The composition and the density `rho` are held fixed. The entropy rises with temperature, and the search runs in `ln T9` between `bounds`; see `density_for_entropy()`.

Parameters
- **entropy** (*float*)
- **abundances** (*Mapping[str, float]*)
- **rho** (*float*)
- **ye** (*float | None*)
- **bounds** (*Tuple[float, float]*)
- **include_partition** (*bool*)
- **electrons** (*bool*)
- **photons** (*bool*)
- **coulomb** (*bool*)

Return type float

nucnetpy.thermo.**constant_entropy_thermo** (*entropy*, *density*, *species*, *ye=None*, *bounds=(0.001, 100.0)*, *include_partition=True*, *electrons=True*, *photons=True*, *coulomb=False*)

Return a thermodynamic callable for a constant-entropy expansion.

`density(t)` gives the density at time `t`. Each time the network asks for the conditions, the temperature is set so that the matter, with its composition at that moment, has the given `entropy` per nucleon. Pass the result as the `thermo` argument of `nucnetpy.evolve_zone()`.

This is the constant-entropy evolution of the NucNet Tools blog. The entropy is held fixed, so the heat released by the reactions is not fed back; that is a good approximation once the burning has slowed down, as in the freeze-out from a hot, expanding plasma. `ye` defaults to the composition's own `sum Z Y` at each call.

The search starts from the previous temperature, so following a smooth expansion costs a few entropy evaluations per call.

Parameters
- **entropy** (*float*)
- **density** (*Callable[[float], float]*)
- **ye** (*float | None*)
- **bounds** (*Tuple[float, float]*)
- **include_partition** (*bool*)
- **electrons** (*bool*)
- **photons** (*bool*)
- **coulomb** (*bool*)

nucnetpy.thermo.**entropy_ideal_ions** (*rho*, *temperature*, *ytot*)

Deprecated: return `Ytot * (5/2 + ln(T^1.5 / rho))`, a scale-free proxy.

This is not the physical entropy: it has no nuclear masses, partition functions or physical constants, and no electrons or photons. For pure helium-4 at `T = 5e9 K` and `rho = 1e8 g /cm^3` it gives 4.39 against a true ion entropy of 3.24. Use `entropy_per_nucleon()` instead. `temperature` is in kelvin.

Parameters
- **rho** (*float*)
- **temperature** (*float*)
- **ytot** (*float*)

Return type float

nucnetpy.thermo.**density_from_entropy** (*entropy*, *temperature*, *ytot*)

Deprecated: invert `entropy_ideal_ions()` for the density.
A physical entropy does not give a physical density here: inverting the true entropy of pure helium-4 at `rho = 1e8 g/cm^3` returns about `1e10`. Use `density_for_entropy()` instead.

Parameters
- **entropy** (*float*)
- **temperature** (*float*)
- **ytot** (*float*)

Return type float

nucnetpy.thermo.**temperature_from_entropy** (*entropy*, *rho*, *ytot*)

Deprecated: invert `entropy_ideal_ions()` for the temperature, in kelvin.
Use `t9_for_entropy()` instead.

Parameters
- **entropy** (*float*)
- **rho** (*float*)
- **ytot** (*float*)

Return type float

nucnetpy.thermo.**sound_speed_ideal_gamma** (*rho*, *pressure*, *gamma=1.6666666666666667*)

Parameters
- **rho** (*float*)
- **pressure** (*float*)
- **gamma** (*float*)

Return type float

nucnetpy.thermo.**q_energy_erg_per_g** (*q_mev*)

Parameters **q_mev** (*float*)

Return type float

The electron-positron gas: chemical potential, pressure, energy, entropy.

NucNet Tools takes the properties of the electrons from libstatmech. This module is the pure-Python counterpart. Given the temperature, the density and the electron fraction `Ye` it finds the electron chemical potential and, from it, everything else the rest of the package needs:

- the pressure, energy and entropy of the electrons and positrons, used by `nucnetpy.thermo.entropy_per_nucleon()`;
- the electrons' contribution to Debye screening, used by `nucnetpy.screening.SkyNetScreening`.

The gas is treated exactly: the electrons may be relativistic and degenerate to any degree, and the positrons that appear in hot, thin matter are included. Their chemical potential is the negative of the electrons' (the reaction `e- + e+ <-> photons` is in equilibrium), and charge neutrality fixes the net number of electrons per nucleon to `Ye`.

Notation. `beta = kT / (m_e c^2)` and `eta` is the electron chemical potential without the rest mass, in units of kT. The positrons then have `eta+ = -eta - 2/beta`. All quantities come from the integrals

I = integral over x of (powers of x and (1 + beta x)) / (exp(x - eta) + 1)

where `x` is the kinetic energy in units of kT. They are evaluated by Gauss-Legendre quadrature in `t = sqrt(x)`, which removes the square-root behaviour at `x = 0`, with the intervals packed around the Fermi edge at `x = eta`. The quadrature agrees with SciPy's adaptive integration to about one part in 10^12 (see the tests).

class nucnetpy.electrons.**ElectronGas** (*t9*, *rho*, *ye*, *eta*, *beta*, *n_electrons*, *n_positrons*, *pressure*, *energy*, *entropy_density*, *dn_deta*)

Bases: `object`

The state of the electron-positron gas at one `(T9, rho, Ye)`.

Densities are in cm^-3, the pressure and energy densities in erg cm^-3 and the entropy density in units of k_B per cm^3. The energies are kinetic energies; the rest mass of the positron pairs is in `pair_rest_energy`.

Parameters

- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)
- **eta** (*float*)
- **beta** (*float*)
- **n_electrons** (*float*)
- **n_positrons** (*float*)
- **pressure** (*float*)
- **energy** (*float*)
- **entropy_density** (*float*)
- **dn_deta** (*float*)

t9: **float**

rho: **float**

ye: **float**

eta: **float**

beta: **float**

n_electrons: **float**

n_positrons: **float**

pressure: **float**

energy: **float**

entropy_density: **float**

dn_deta: **float**

property mu_mev: float
Electron chemical potential in MeV, including the rest mass.

property pair_rest_energy: float
Rest energy of the positrons and the electrons paired with them, erg cm^-3.

property entropy_per_nucleon: float
Entropy of the electrons and positrons per nucleon, in units of k_B.

property degeneracy_factor: float
How strongly the electrons screen compared with a classical gas.
`(d n / d eta) / (n- - n+)`: one for a non-degenerate electron gas, falling towards zero as the electrons become degenerate, and above one when positron pairs add to the screening.

property screening_term: float
The electrons' share of the Debye sum, per nucleon.
Debye screening by the ions is `sum Z^2 Y`; the electrons and positrons add `kT dn/dmu / (rho N_A)`, which is this value. It equals `Ye` for non-degenerate electrons and tends to zero for degenerate ones.

nucnetpy.electrons.**electron_chemical_potential** (*t9*, *rho*, *ye*)
Return `eta`, the electron chemical potential over kT without rest mass.
It is the value for which the net electron density `n- - n+` equals `rho N_A Ye`.

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type float

nucnetpy.electrons.**electron_gas** (*t9*, *rho*, *ye*)
Return the full `ElectronGas` state at `(t9, rho, ye)`.
`t9` is the temperature in 10^9 K, `rho` the density in g cm^-3 and `ye` the net number of electrons per nucleon.

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float*)

Return type *ElectronGas*

Validation and regression utilities for replacing NucNet Tools workflows.

class nucnetpy.validation.**ValidationIssue** (*level: 'str', code: 'str', message: 'str'*)
Bases: `object`

Parameters
- **level** (*str*)
- **code** (*str*)
- **message** (*str*)

level: str

code: str

message: str

nucnetpy.validation.**validate_network** (*network*, *strict=False*)

Parameters
- **network** (*Network*)
- **strict** (*bool*)

Return type *List*[*ValidationIssue*]

nucnetpy.validation.**validate_zone** (*zone*, *network=None*, *atol=1e-06*)

Parameters
- **zone** (*Zone*)
- **network** (*Network | None*)
- **atol** (*float*)

Return type *List*[*ValidationIssue*]

nucnetpy.validation.**compare_abundances** (*a*, *b*, *rtol=1e-06*, *atol=1e-30*)

Parameters
- **a** (*Mapping[str, float]*)
- **b** (*Mapping[str, float]*)
- **rtol** (*float*)
- **atol** (*float*)

Return type *Dict*[str, float]

nucnetpy.validation.**max_relative_abundance_error** (*a*, *b*, *atol=1e-30*)

Parameters
- **a** (*Mapping[str, float]*)
- **b** (*Mapping[str, float]*)
- **atol** (*float*)

Return type float

nucnetpy.validation.**regression_summary** (*reference*, *candidate*)

Parameters
- **reference** (*Mapping[str, float]*)
- **candidate** (*Mapping[str, float]*)

Return type *Dict*[str, object]

Pure-Python rate modification hooks replacing user/rate_modifiers.*.

class nucnetpy.rate_modifiers.**RateModifierRegistry** (*modifiers: 'Dict[str, RateModifier]'=<factory>*)

Bases: object

Parameters **modifiers** (*Dict[str, Callable[[object, float, float, float | None, Mapping[str, float]], float]]*)

modifiers: Dict[str, Callable[[object, float, float, float | None, Mapping[str, float]], float]]

register (*key*, *func*)

Parameters
- **key** (*str*)
- **func** (*Callable[[object, float, float, float | None,*

Mapping[str, float]], float])

Return type None

factor (*reaction*, *t9*, *rho*, *ye*, *abundances*)

Parameters
- **t9** (*float*)
- **rho** (*float*)
- **ye** (*float | None*)
- **abundances** (*Mapping[str, float]*)

Return type float

nucnetpy.rate_modifiers.**constant_factor** (*value*)

Parameters **value** (*float*)

Return type *Callable*[[object, float, float, float | None, *Mapping*[str, float]], float]

nucnetpy.rate_modifiers.**exp_temperature_factor** (*alpha*, *power=1.0*)

Parameters
- **alpha** (*float*)
- **power** (*float*)

Return type *Callable*[[object, float, float, float | None, *Mapping*[str, float]], float]

Graph export helpers replacing the Graphviz-oriented examples.

nucnetpy.graph.**reaction_network_dot** (*network*, *min_rate=0.0*, *t9=1.0*, *rho=1.0*)

Parameters
- **network** (*Network*)
- **min_rate** (*float*)
- **t9** (*float*)
- **rho** (*float*)

Return type str

Physical constants used by nucnetpy.

The values are SI/cgs mixed in the conventional astrophysical style used by network codes. They are intentionally centralized so advanced users can swap or audit constants when comparing with libnucnet/NucNet Tools builds.

Chapter 7

Command line

Installing the package puts a `nucnetpy` command on the path. This page is generated from the argument parser by `docs/manual/generate_cli_reference.py`, so it matches the code.

```
usage: nucnetpy [-h]

{summary,print-output,largest-x,zone-abundances,zone-properties,element-abundances,rates,flows,y
...

Pure-Python NucNet Tools replacement commands

positional arguments:

{summary,print-output,largest-x,zone-abundances,zone-properties,element-abundances,rates,flows,y
net-flows           forward, detailed-balance reverse, and net fluxes
    charge-flows        per-reaction dYe/dt contributions
    timescales          shortest species timescales Y/|dY/dt|
    entropy-generation  dS/dt per nucleon in k_B/s
    qse                 constrained cluster equilibrium (libnuceq)
    jina-summary        summarise separate JINA nuclide and reaction XML files
    jina-combine        combine JINA nuclide and reaction XML files into one
                        nucnetpy XML file

options:
  -h, --help            show this help message and exit
```

7.1 summary

```
usage: nucnetpy summary [-h] xml

positional arguments:
  xml

options:
  -h, --help  show this help message and exit
```

7.2 print-output

```
usage: nucnetpy print-output [-h] [--max-zones MAX_ZONES]
                             [--min-abundance MIN_ABUNDANCE]
                             xml
```

```
positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --max-zones MAX_ZONES
  --min-abundance MIN_ABUNDANCE
```

7.3 largest-x

```
usage: nucnetpy largest-x [-h] [-n N] [--zone-index ZONE_INDEX]
                          [--min-x MIN_X]
                          xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  -n N
  --zone-index ZONE_INDEX
  --min-x MIN_X
```

7.4 zone-abundances

```
usage: nucnetpy zone-abundances [-h] [--zone-index ZONE_INDEX]
                                [--min-abundance MIN_ABUNDANCE]
                                xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --min-abundance MIN_ABUNDANCE
```

7.5 zone-properties

```
usage: nucnetpy zone-properties [-h] [--zone-index ZONE_INDEX] xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
```

7.6 element-abundances

```
usage: nucnetpy element-abundances [-h] [--zone-index ZONE_INDEX] xml element

positional arguments:
  xml
  element

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
```

7.7 rates

```
usage: nucnetpy rates [-h] [--t9 T9] [--rho RHO] [--min-rate MIN_RATE] xml

positional arguments:
  xml

options:
  -h, --help           show this help message and exit
  --t9 T9
  --rho RHO
  --min-rate MIN_RATE
```

7.8 flows

```
usage: nucnetpy flows [-h] [--zone-index ZONE_INDEX] [--t9 T9] [--rho RHO]
                      [--min-flow MIN_FLOW]
                      xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t9 T9
  --rho RHO
  --min-flow MIN_FLOW
```

7.9 ydot

```
usage: nucnetpy ydot [-h] [--zone-index ZONE_INDEX] [--t9 T9] [--rho RHO]
                     [--min-abs MIN_ABS]
                     xml

positional arguments:
  xml

options:
```

```
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t9 T9
  --rho RHO
  --min-abs MIN_ABS
```

7.10 net-flows

```
usage: nucnetpy net-flows [-h] [--zone-index ZONE_INDEX] [--t9 T9] [--rho RHO]
                          [--min-flow MIN_FLOW]
                          xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t9 T9
  --rho RHO
  --min-flow MIN_FLOW
```

7.11 charge-flows

```
usage: nucnetpy charge-flows [-h] [--zone-index ZONE_INDEX] [--t9 T9]
                             [--rho RHO] [--min-flow MIN_FLOW]
                             xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t9 T9
  --rho RHO
  --min-flow MIN_FLOW
```

7.12 timescales

```
usage: nucnetpy timescales [-h] [--zone-index ZONE_INDEX] [--t9 T9]
                           [--rho RHO] [-n N]
                           xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t9 T9
  --rho RHO
  -n N
```

7.13 entropy-generation

```
usage: nucnetpy entropy-generation [-h] [--zone-index ZONE_INDEX] [--t9 T9]
                                   [--rho RHO]
                                   xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t9 T9
  --rho RHO
```

7.14 conservation

```
usage: nucnetpy conservation [-h] [--max MAX] xml

positional arguments:
  xml

options:
  -h, --help  show this help message and exit
  --max MAX
```

7.15 remove-duplicates

```
usage: nucnetpy remove-duplicates [-h] xml output

positional arguments:
  xml
  output

options:
  -h, --help  show this help message and exit
```

7.16 remove-invalid

```
usage: nucnetpy remove-invalid [-h] xml output

positional arguments:
  xml
  output

options:
  -h, --help  show this help message and exit
```

7.17 export-zone-xml

```
usage: nucnetpy export-zone-xml [-h] [--zone-index ZONE_INDEX] xml output

positional arguments:
  xml
  output

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
```

7.18 reactions-latex

```
usage: nucnetpy reactions-latex [-h] xml output

positional arguments:
  xml
  output

options:
  -h, --help  show this help message and exit
```

7.19 species-history

```
usage: nucnetpy species-history [-h] xml species

positional arguments:
  xml
  species

options:
  -h, --help  show this help message and exit
```

7.20 evolve-zone

```
usage: nucnetpy evolve-zone [-h] [--zone-index ZONE_INDEX] [--t0 T0] [--t1 T1]
                            [--steps STEPS] [--t9 T9] [--rho RHO]
                            [--method
{bdf,radau,lsoda,rk45,dop853,adaptive_implicit_euler,rk4,euler,implicit_euler,backward_euler}]
[--log-time] [--min-abundance MIN_ABUNDANCE]
                            xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t0 T0
  --t1 T1
```

```
  --steps STEPS
  --t9 T9
  --rho RHO
  --method
{bdf,radau,lsoda,rk45,dop853,adaptive_implicit_euler,rk4,euler,implicit_euler,backward_euler}
--log-time
  --min-abundance MIN_ABUNDANCE
```

7.21 energy-generation

```
usage: nucnetpy energy-generation [-h] [--zone-index ZONE_INDEX] [--t9 T9]
                                  [--rho RHO]
                                  xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --zone-index ZONE_INDEX
  --t9 T9
  --rho RHO
```

7.22 net-dot

```
usage: nucnetpy net-dot [-h] [-o OUTPUT] [--t9 T9] [--rho RHO]
                        [--min-rate MIN_RATE]
                        xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  -o OUTPUT, --output OUTPUT
  --t9 T9
  --rho RHO
  --min-rate MIN_RATE
```

7.23 validate

```
usage: nucnetpy validate [-h] [--strict] [--max MAX] [--max-zones MAX_ZONES]
                         xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --strict
  --max MAX
  --max-zones MAX_ZONES
```

7.24 nse

```
usage: nucnetpy nse [-h] --t9 T9 --rho RHO --ye YE [--coulomb]
                    [--min-abundance MIN_ABUNDANCE] [--min-x MIN_X]
                    xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --t9 T9
  --rho RHO
  --ye YE
  --coulomb             apply Bravo & Garcia-Senz Coulomb corrections
  --min-abundance MIN_ABUNDANCE
  --min-x MIN_X
```

7.25 qse

```
usage: nucnetpy qse [-h] --t9 T9 --rho RHO --ye YE [--cluster SP1,SP2,...:Y]
                    [--coulomb] [--min-abundance MIN_ABUNDANCE]
                    [--min-x MIN_X]
                    xml

positional arguments:
  xml

options:
  -h, --help            show this help message and exit
  --t9 T9
  --rho RHO
  --ye YE
  --cluster SP1,SP2,...:Y
                        cluster species and constrained total abundance;
                        repeatable
  --coulomb
  --min-abundance MIN_ABUNDANCE
  --min-x MIN_X
```

7.26 jina-summary

```
usage: nucnetpy jina-summary [-h] [--zones-xml ZONES_XML] [--show-invalid]
                             [--max-invalid MAX_INVALID]
                             nuclides_xml reactions_xml

positional arguments:
  nuclides_xml
  reactions_xml

options:
  -h, --help            show this help message and exit
  --zones-xml ZONES_XML
  --show-invalid
```

```
  --max-invalid MAX_INVALID
```

7.27 jina-combine

```
usage: nucnetpy jina-combine [-h] [--zones-xml ZONES_XML]
                             nuclides_xml reactions_xml output_xml

positional arguments:
  nuclides_xml
  reactions_xml
  output_xml

options:
  -h, --help            show this help message and exit
  --zones-xml ZONES_XML
```

Conversion map from original NucNet Tools concepts to `nucnetpy`

Original area	Python replacement
`Libnucnet__Nuc` / species handling	`nucnetpy.species.Species`, `Network.species`
`Libnucnet__Reac` / reactions	`nucnetpy.reactions.Reaction`, `ReactionNetwork`, `RateFit`, `TabularRate`
ReacLib seven-parameter rates	`RateFit.rate(t9)`
Reaction duplicate/invalid utilities	`ReactionNetwork.remove_duplicates`, `filter_valid`, CLI `remove-duplicates`, `remove-invalid`
Zone XML and abundance output	`Zone`, `Network.zones`, `nucnetpy.io.xml`
Analysis examples (`print_output`, largest X, flows, ydot)	`nucnetpy.analysis` and CLI commands
Graph utilities	`nucnetpy.graph.reaction_network_dot`, CLI `net-dot`
Screening user callbacks	`nucnetpy.screening` with weak/intermediate models and solver callback hook
Weak-rate utilities	`nucnetpy.weak.WeakRateTable`, `read_weak_table`, solver weak-rate hook
`libnuceq`/NSE workflows	`nucnetpy.nse.solve_nse`, `equilibrium_ratio`
Matrix/Jacobian solver utilities	`nucnetpy.solver.jacobian`, `jacobian_sparsity`, SciPy BDF/Radau/LSODA
Multi-zone evolution	`evolve_network_zones`
HDF5 helper workflow	`nucnetpy.io.hdf5`

8.1 What is implemented

- Flexible XML parsing for common legacy structures and compact Python-written XML.
- Partition functions and optional properties are preserved.
- ReacLib, tabular, and constant rates are supported.
- NSE solves are available for network-level equilibrium initialization.
- Screening and weak-rate hooks are now solver-compatible.
- A sparse numerical Jacobian and sparsity pattern are available for stiff integration.
- Validation/regression helpers are included.

8.2 What still needs project-specific validation

A full replacement can now be used and extended, but exact reproduction of a historical C++ run requires comparing against your original input/output files. The main tunable points are:

1. exact XML schema variants used by your input files,
2. original mass excess and partition-function data,
3. exact screening callback selected in the original C++ build,
4. weak-rate table source and interpolation convention,
5. solver tolerance, timestep, and rate-modifier settings.

Chapter 9

Blog workflow coverage

Mapping of the science workflows from Bradley Meyer's NucNet Tools blog (https://sourceforge.net/u/mbradle/blog/, 94 posts, 2012–2019) to nucnetpy features. Installation/infrastructure posts (Cygwin, Boost, compilers, AWS, SourceForge housekeeping — 24 posts) do not apply to a pure-Python package and are omitted.

9.1 Covered

Blog workflow	nucnetpy feature
Running a first network calculation; changing input conditions; dynamic H burning	`evolve_zone`, `constant_thermo`, CLI `evolve-zone`
Analyzing a network calculation; printing properties and mass fractions	`analysis`, CLI `print-output`, `largest-x`, `zone-*`
Understanding/updating/validating nuclear and reaction data	`io.xml`, `io.jina`, `validate_network`, CLI `validate`
Merging nuclear and reaction data	`combine_jina_xml`, CLI `jina-combine`
Calculating reaction rates	`RateFit`, `TabularRate`, CLI `rates`
Comparing forward and reverse reaction rates	**`reverse_rate`, `reverse_reaction` (detailed balance)**
Computing reaction flows (forward + reverse)	**`net_flows`, CLI `net-flows`**; forward-only: `flows`
Constructing the rate equations	`solver.rhs`, `jacobian`, `jacobian_sparsity`
Finding valid reactions; conservation	`invalid_reactions`, `conserves_a_z`, CLI `conservation`
Modifying reaction rates during a calculation; user-defined rate modifiers	`rate_modifiers`
Specifying an abundance in a network calculation	`Zone.set_abundance`
Selecting an input network; limiting the reaction network	**`select_species` (new)**, `limit_network`
Nuclide XPath / XSLT node-set extraction	**`select_species` (Z/A/element selection)**
Sorting species in a network calculation	`Network.species_names` (Z, A order)
Calculating nuclear statistical equilibrium	`solve_nse`, CLI `nse`

Comparing network calculations to equilibrium	`solve_nse` + `regression_summary`
Studying the nuclear partition function	`Species.partition`, partition-aware NSE
Understanding abundance moments	`abundance_moment`
Computing the number of heavy nuclei	**`heavy_nuclei_abundance` (new)**
Computing system timescales	**`system_timescales` (new), CLI `timescales`**
Computing charge-changing flows	**`charge_changing_flows` (new), CLI `charge-flows`**
Computing the s-process neutron exposure	**`neutron_exposure` (new)**
Plotting abundances versus nucleon number	`abundances_vs_nucleon_number`
Plotting isotopic abundances versus time	`species_history`, `EvolutionResult.mass_fraction_history`
Computing separation energies	**`separation_energy` (proton branch fixed)**
Computing/understanding the entropy generation rate (incl. weak decay)	**`entropy_generation_rate` (new), CLI `entropy-generation`**
Computing thermodynamic quantities	`thermo`, `electrons` (electron-positron gas: chemical potential, pressure, energy, entropy)
Defining your own thermodynamic (trajectory) function	`ThermoFunction` callables, `hydro.Trajectory`
Using a thermodynamic trajectory file	`read_trajectory`
Running a network calculation with simple hydrodynamics	`exponential_expansion`
Including electron screening; supplying your own screening function	`screening`, solver `screening=` hook
Adding fission to an r-process calculation	**`fission_reaction` (new)**
Running an r-process calculation	`evolve_zone` + JINA XML network
(n,γ)-(γ,n) equilibrium studies	**`reverse_reaction`** to build photodisintegration partners; `solve_nse`
Creating flow diagrams (colours, scalings, currents)	`graph.reaction_network_dot`, CLI `net-dot` (basic)
Creating webnucleo reaction XML from text with Python	`read_reaclib_text`, `io.text`, `write_xml`
Using zone properties; global strings	`Zone.properties`, `Network.metadata`
Selecting a zone as input for a new calculation	CLI `export-zone-xml`
Controlling the frequency of writing output	`times` grid passed to `evolve_zone`
Using wnutils to analyse output	`analysis` module + `io.hdf5`
Weak decays / entropy for weak decay	`weak` tables + `charge_changing_flows`
Updating mass excesses in a nuclide XML	edit `Species.mass_excess`, re-`write_xml`

Counting matrix elements; arrow matrix storage	`jacobian_sparsity` (SciPy sparse replaces arrow solver)

9.2 Ported from the C++ source (r647)

These items were verified against the actual NucNet Tools SVN trunk (`nucnet-tools-code` r647, downloaded from SourceForge) and ported from the named C++ files:

Blog workflow	C++ origin	nucnetpy feature
Studying Coulomb corrections to NSE; including NSE corrections in a network calculation	`user/nse_corr.cpp` (Bravo & García-Senz 1999, Ogata & Ichimaru constants)	`coulomb` module: `gamma_e`, `species_coulomb_chemical_potential_energy` / `_entropy`, `coulomb_nse_correction`; `solve_nse(..., nse_correction=...)` hook
Computing NSE corrections to the entropy	`user/nse_corr.cpp` (`species_coulomb_entropy`)	`coulomb_entropy_per_nucleon`
Comparing network calculations to equilibrium; (n,γ)-(γ,n) equilibrium; constraining the equilibrium	libnuceq clusters via `examples/analysis/compare_equil.cpp`	`solve_qse` with `QSECluster` constraints (one multiplier per cluster on top of μ_p/μ_n)
Calculating cluster flows	`examples/analysis/compute_Ycdot.cpp`	`cluster_ydot`, `cluster_abundance`

Creating/analyzing integrated currents diagrams	`user/flow_utilities.cpp` (`update_flow_currents`: current += (f−r)·dt)	`integrated_currents` over an `EvolutionResult`
Computing the entropy generation rate (per reaction, with reverse flows)	`user/flow_utilities.cpp` (`compute_entropy_generation_rate`)	`entropy_generation_rate(use_reverse=True)` — Σ(f−r)·ln(f/r), non-negative, vanishes at NSE; `reaction_entropy_changes`

9.3 Not ported (by design or still open)

Blog workflow	Status
Inverting the entropy to find the density	`entropy_per_nucleon` (ions with masses, spins and partition functions; electrons and positrons from `nucnetpy.electrons`; photons; optional Coulomb term), `density_for_entropy`, `t9_for_entropy`. The old proxies `density_from_entropy` and `temperature_from_entropy` are deprecated
Constant-entropy evolution with self-consistent entropy inverter	`constant_entropy_thermo(entropy, density, network)` — a `thermo` callable that sets T9 from the current composition at each step; the reactions' heat is not fed back
Electron screening in non-degenerate matter with `SkyNetScreening`	the electron and positron term of the Debye sum is computed from the electron gas (`ElectronGas.screening_term`); `pair_term` overrides it
Electron/positron chemical-potential terms in entropy generation (libstatmech)	open — weak reactions carry tabulated rates instead
Fission *cycling* studies (automatic fragment distributions)	partial — `fission_reaction` builds channels; distributions are user input
Downloading webnucleo XML files	use the URLs in the blog post; nucnetpy reads them directly
XML inclusion (XInclude) / XSLT transforms	use `lxml` externally, then `read_xml`

Chapter 10

Pure Python port status

The C++ source contains helper files under `nnt/` and `user/`. NucNetPy provides Python equivalents for the practical computational layers rather than a wrapper around a compiled backend.

Original area	Python module	Status
`nnt/math.*`	`nucnetpy.mathutils`	Ported: linear, bilinear, 2-D interpolation behaviour
wrappers/containers	`core.py`, `species.py`	Ported core containers
network/reaction utilities	`reactions.py`, `validation.py`	Ported main rate/flow/conservation APIs
`screen.*`	`screening.py`	Pure-Python weak and intermediate screening; exact formula variants still need golden validation
`weak_utilities.*`, `two_d_weak_rates.*`	`weak.py`	Ported 2-D weak-rate table interpolation and weak source terms
`thermo.*`	`thermo.py`, `electrons.py`, `nse.py`	Entropy of ions, electrons and positrons, and photons; density and temperature from entropy; NSE. The electron gas is checked against closed-form limits and thermodynamic identities, not against a libstatmech build
`matrix_solver.*`, `ilu_solvers.*`	`matrix_solver.py`, `solver.py`	Ported to SciPy sparse solvers and stiff integrators
`hydro.*`, `hydro_helper.*`	`hydro.py`	Ported trajectory interpolation and exponential expansion

network_limiter.*	network_limiter.py	Ported species/reaction subnetwork limiting
nuclear_decay_utilities.*	decay.py	Ported half-life/decay record tools
neutrino_rate_functions.*	neutrino.py	Ported neutrino luminosity/cross-section helpers
hdf5_routines.*	io/hdf5.py	Ported compact HDF5 save/load
graph.h	graph.py	Ported DOT export

10.1 What remains for strict numerical identity

No C++ is used in this package. To make the Python code exactly match a chosen original C++ build, add golden-output tests generated once from the original tools, then tune the pure-Python formulas and tolerances until those tests pass.

The framework for this now exists in-repo. tests/test_golden_identity.py implements the full recommended order below against golden files in tests/golden/, and validation/generate_golden.py regenerates those files:

1. XML round-trip tests for your real networks — test_xml_round_trip, test_round_trip_is_a_fixed_point (identity tests, no golden data needed).
2. Rate-by-rate ReacLib comparisons on a T9 grid — test_reaclib_rates_on_t9_grid vs rates_reaclib.json.
3. Screening-factor comparisons for your chosen user/screen.cpp model — test_screening_factors vs screening.json.
4. Weak-rate table comparisons — test_weak_rate_table_interpolation vs weak_rates.json.
5. One-zone RHS ydot comparisons at fixed state — test_ydot_at_fixed_states vs ydot.json (also asserts nucleon-number conservation of the RHS).
6. Full trajectory comparisons with identical time grids and tolerances — test_full_trajectory vs trajectory.json (deterministic fixed-step rk4), plus a solver cross-check that adaptive BDF agrees with the rk4 path.

Out of the box the goldens are self-consistent snapshots of the current Python numerics, so any future change to a formula fails the suite immediately. To validate against a specific C++ NucNet Tools build:

1. Replace the fixture inputs (tests/golden/golden_network.xml, golden_weak_rates.txt) with your production data, or keep them and reproduce the same inputs on the C++ side.
2. Run the same grids through the original tools and write their outputs into the data blocks of the JSON files (set source to your build id).
3. Loosen each file's rtol/atol to the agreement you require — the tolerances are read from the golden files, not hard-coded in the tests.

Exact numerical identity policy

NucNetPy is pure Python. It contains no C++ code, does not call the original NucNet Tools programs, and has no "exact backend". It therefore cannot promise bit-for-bit agreement with a particular C++ build on its own.

11.1 Why identity cannot simply be assumed

Two codes that implement the same equations can still disagree in the last digits, or by more, because results depend on:

- the nuclear data: mass excesses, partition-function tables and their interpolation, and the rate fits;
- modelling choices: the screening prescription, how reverse rates are formed, and which species enter an equilibrium solve;
- numerical choices: the integrator, its tolerances and time grid, the order in which reactions are summed, and the linear-algebra library;
- the platform: compiler, optimisation flags and floating-point rounding.

Agreement has to be shown by comparison, not asserted.

11.2 How to compare with a C++ build you run yourself

The repository carries the machinery for this; the port status notes (`docs/PURE_PYTHON_PORT_STATUS.md`) describe it in detail.

1. Run the original NucNet Tools on the inputs in `tests/golden/` (or replace those inputs with your own network and run both codes on them).
2. Write the C++ outputs into the `data` blocks of the JSON files in `tests/golden/`, and set each file's `source` to a label for your build.
3. Set each file's `rtol` and `atol` to the agreement you need. The tests read the tolerances from the files, so no test code has to change.
4. Run `pytest tests/test_golden_identity.py`. A failure names the quantity, the value from each code and the tolerance.

Out of the box those files hold snapshots of NucNetPy's own output, so the same tests also catch any unintended change to the Python numerics. `validation/generate_golden.py` regenerates the snapshots after an intended change.

11.3 Development rule

When a C++ example is converted to Python, add a golden-output test for it:

1. Run the original C++ program on a fixed input file and save its output.
2. Run the Python equivalent on the same input.
3. Compare with zero tolerance first. If the platforms differ in the last bits, record the tolerance you need and why.

Only modules that pass such tests should be described as exact replacements.

Chapter 12

JINA XML workflow in nucnetpy

The original C++ NucNet Tools examples usually begin from JINA/libnucnet XML files. The Python port follows the same idea, but uses a Python object model.

12.1 Files

Use your stored XML files as follows:

- `jina_nuclides.xml`: nuclides/species, including Z, A, mass excess, spin, and partition data.
- `jina_reactions.xml`: reactions and ReacLib rate coefficients.
- optional `zones.xml`: initial abundances or output zones.

12.2 Python usage

```
from nucnetpy import read_jina_xml

net = read_jina_xml("jina_nuclides.xml", "jina_reactions.xml")
print(net.validate())
```

With zones:

```
net = read_jina_xml(
    "jina_nuclides.xml",
    "jina_reactions.xml",
    zones_xml="initial_zones.xml",
)
```

12.3 Command-line usage

```
nucnetpy jina-summary jina_nuclides.xml jina_reactions.xml
nucnetpy jina-combine jina_nuclides.xml jina_reactions.xml combined_network.xml
nucnetpy rates combined_network.xml --t9 1.0 --rho 1e5
nucnetpy conservation combined_network.xml
```

12.4 Important numerical point

To approach numerical identity with the original C++ tools, use the exact same JINA XML files, do

not reorder or filter reactions unless intentionally testing that operation, and compare rates/flows at fixed `T9`, `rho`, and abundance values.

Where to start

If you have never used the package, read Installation and Concepts, then work through the tutorial notebooks in `notebooks/`, which are the fastest way in.

If you are migrating an existing NucNet Tools workflow, Conversion map from original NucNet Tools concepts to nucnetpy maps the original concepts onto this API and Blog workflow coverage maps each blog workflow onto the feature that replaces it.

If you are about to run a production calculation, read Pitfalls first. It describes two behaviours that produce plausible but wrong answers without raising an error.

Chapter 14

Citing

Please cite the archived release (10.5281/zenodo.20756798) and acknowledge the NucNet Tools and libnucnet ecosystem, whose data model and workflows this package reimplements, together with the JINA reaction-rate database that supplies the nuclear data.

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